PHYSICS

1)

The ranges and heights for two projectiles projected with the same initial speed at angles 42° and 48° with the horizontal are R_1, R_2 and H_1, H_2 respectively. Choose the correct option :

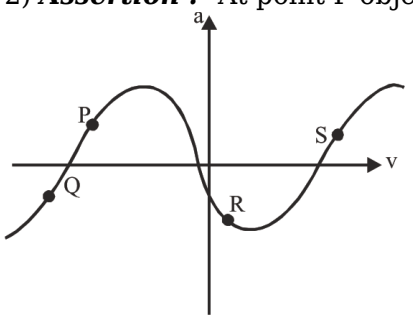
(1) $R_1 > R_2$ and $H_1 = H_2$

(2) $R_1 = R_2$ and $H_1 < H_2$

(3) $R_1 < R_2$ and $H_1 < H_2$

(4) $R_1 = R_2$ and $H_1 = H_2$

2) ***Assertion :-*** At point P object is slowing down.



Reason :- If acceleration is positive, object must speed up.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

3) A ball is projected from a certain point on the surface of a planet at a certain angle with the horizontal surface. The horizontal and vertical displacement x and y vary with time t in second as $x = 10\sqrt{3}t$ and $y = 10t - t^2$. The maximum height attained by the ball is :

(1) 100 m

(2) 75 m

(3) 50 m

(4) 25 m

4) A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product t_1t_2 is :

(1) R/g

(2) R/4g

(3) 2R/g

(4) R/2g

5) During a cycle race, two persons happen to cross a point x simultaneously with speeds 16 m/sec and 12 m/sec respectively. In order to win the race they increase their speeds at the rate 1 m/s^2 and 2 m/s^2 respectively. Unfortunately both reach the final point y at the same instant of time. Find the distance xy :-

(1) xy = 120 m

(2) xy = 160 m

(3) xy = 100 m

(4) xy = 80 m

6) The position of a particle moving in a straight line is described by the relation $S = 6 + 12t - 2t^2$. Here S is in meters and t is in second. The distance covered by the particle in first 5sec is :-

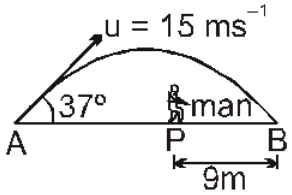
(1) 20m

(2) 32m

(3) 24m

(4) 26m

7) A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that



height of man is negligible in comparison to maximum height of projectile.

(1) 3 ms^{-1}

(2) 5 ms^{-1}

(3) 9 ms^{-1}

(4) 12 ms^{-1}

8) If $x = 1 + 2t^3 + 3t^4$ find change in velocity between t = 1 second to t = 2 seconds

(1) 102 m/s

(2) 82 m/s

(3) 122 m/s

(4) 92 m/s

9) In a straight line motion particle start from rest and travel with constant acceleration covers distance x in first 3 seconds and y in next 9 seconds then ratio $\frac{y}{x}$ equals :-

(1) 1 : 15

(2) 1 : 9

(3) 9 : 1

(4) 15 : 1

10) Drops of water are falling from roof at equal time interval, when 5th drop is about to fall 1st drop reaches the ground. Then find distance between 2nd and 4th drop if total height of roof is 48m.

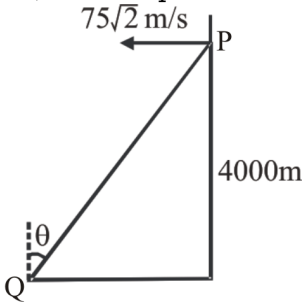
(1) 8 m

(2) 3 m

(3) 24 m

(4) 36 m

11) An aeroplane is travelling horizontally at a height of 4000 m from the ground.



The aeroplane, when at a point P, drops a bomb to hit a stationary target Q on the ground. In order that the bomb hits the target, what angle θ must the line PQ make with the vertical ? [g = 10 m/s^2]

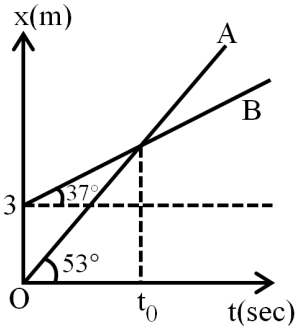
(1) 37°

(2) 53°

(3) 45°

(4) 30°

12) For given curve for particle A and B select incorrect statement.



(1) $V_A > V_B$

(2) $t_o = \frac{36}{7}$ sec

(3) Acceleration of both particles are zero.

(4) Relative velocity of particles is zero.

13) A boat having a speed of 5 km/h in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes. The speed of the river in km/h is:

(1) 1

(2) 3

(3) 4

(4) $\sqrt{4}$

14)

A particle is moving with uniform acceleration along a straight line . Its velocities at A & B are respectively 7 m/s & 17 m/s . M is mid point of AB . If t_1 is the time taken to go from A to M and t_2 the time taken to go from M to B, the ratio $\frac{t_1}{t_2}$ is equal to :-

(1) 3 : 2

(2) 3 : 1

(3) 2 : 1

(4) 2 : 3

15) Choose the correct options :
(a) a scalar quantity can never take negative values
(b) a scalar quantity have same value for observer with different orientations of the axes
(c) Every quantity which has magnitude & direction is always vector
(d) If the resultant of several vectors is zero, then the y-component of resultant vector must be zero

(1) a, b, c and d are correct

(2) a, b & c are correct

(3) a & c are correct

(4) b & d are correct

16) The position vector $\vec{A}$ varies with time as $\vec{A} = t\hat{i} + 2t^2\hat{j}$. It's rate of change of angle (per second) with +x axis at t = $\frac{1}{2}$ sec. is :

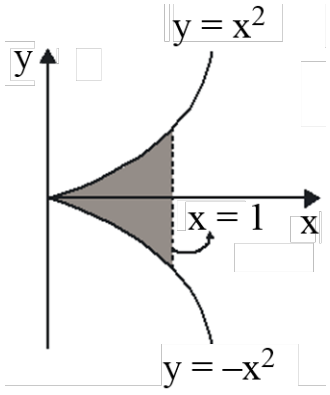
(1) $\frac{1}{2}$

(2) 2

(3) 1

(4) 4

17) If the shaded area is $\frac{x}{6}$ then x is :



(1) 1

(2) 2

(3) 3

(4) 4

18) If in a new hypothetical system of units, the new unit of mass is 5 kg, new unit of length is 10 metre and new unit of time is 20 seconds, choose the correct statement(s).
(a) 1 SI unit of velocity is equal to 2 new units of velocity.
(b) 1 SI unit of power is equal to 16 new units of power.
(c) 1 SI unit of energy is equal to 54 new unit of energy.
(d) 1 SI unit of force is equal to 10 new units of force.

(1) a & b

(2) c & d

(3) a & c

(4) a & d

19) Assume that the largest stone of mass m that can be moved by a flowing river depends upon the velocity of flow (v), density (d) and the acceleration due to gravity (g). If 'm' varies as the kth power of acceleration due to gravity. Then find the value of |2K|.

(1) 3

(2) $\frac{3}{2}$

(3) 6

(4) 4

20) The speed of a wave produced in water is given by $\upsilon = \lambda^a g^b \rho^c$. Where λ, g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a, b and c respectively, are :

(1) $\frac{1}{2}, \frac{1}{2}, 0$

(2) 1, 1, 0

(3) 1, –1, 0

(4) $\frac{1}{2}, 0, \frac{1}{2}$

21) Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**
Assertion A : A spherical body of radius (5 ± 0.1) mm having a particular density is falling through a liquid of constant density. The percentage error in the calculation of its terminal velocity is 4%.
Reason R : The terminal velocity of the spherical body falling through the liquid is inversely proportional to its radius.
In the light of the above statements, choose the ***correct*** answer from the options given below.

(1) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

(2) Both **A** and **R** are true and **R** is the correct explanation of **A**

(3) **A** is false but **R** is true

(4) **A** is true but **R** is false

22) The main scale of a vernier caliper reads in mm and its Vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. The reading for the given situation will be :-

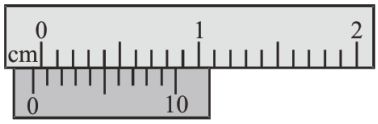
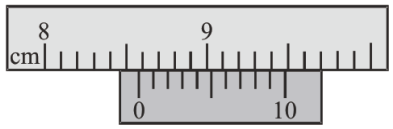
Fig 1 

Fig 2 

(1) 7.58 cm

(2) 8.79 cm

(3) 8.63 cm

(4) 9.78 cm

23) Match the physical quantities mentioned in **Column I** with their respective dimensional given in **Column II.**

	Column I		**Column II**
A	Inductance (L)	P	$[M^1L^2T^{-2}k^{-1}]$
B	Boltzmann's constant (k)	Q	$M^1L^3T^{-3}A^{-2}$
C	Magnetic permeability (μ_0)	R	$[M^1L^2T^{-2}\ A^{-2}]$
D	Resistivity (ρ)	S	$[M^1L^1T^{-2}\ A^{-2}]$

(1) A-Q, B-S, C-P, D-R

(2) A-S, B-P, C-Q, D-R

(3) A-R, B-P, C-S, D-Q

(4) A-Q, B-P, C-R, D-Q

24) A vernier calliper has a zero error of +0.2 mm. For a tube both internal and external diameters are measured and then subtracted, then in result, zero error will.

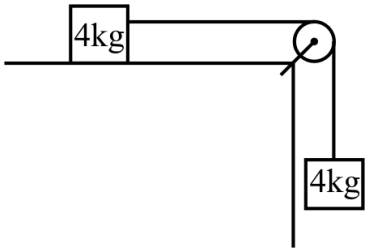
(1) remain same

(2) become two times

(3) cancels out

(4) cannot be predicted

25) In the given arrangement, there is no friction anywhere. Pulley and string are ideal. Acceleration due to gravity is 10 m/s^2 downward. Find out total work done by gravity in 2 sec, if system starts



from rest.

(1) 100 J
(2) 400 J
(3) 200 J
(4) 800 J

26) If the momentum of a body is increased by 200 % then increase in its kinetic energy will be :-

(1) 200%
(2) 600%
(3) 800%
(4) 100%

27) If $F = 2x^2 - 3x - 2$, then choose correct option :-

(1) $x = -1/2$ is position of stable equilibrium
(2) $x = 2$ is position of stable equilibrium
(3) $x = -1/2$ is position of unstable equilibrium
(4) $x = 2$ is position of neutral equilibrium

28) A particle of mass 100 g is thrown vertically upwards with a speed of 5 m/s. During the time when the particle reaches the maximum height, the work done by the force of gravity is :-

(1) –0.5 J
(2) –1.25 J
(3) 1.25 J
(4) 0.5 J

29)

Two springs of spring constants 1500 N/m and 3000 N/m respectively are stretched with the same force. They will have potential energy in the ratio

(1) 4 : 1
(2) 1 : 4
(3) 2 : 1
(4) 1 : 2

30) The power of a water pump is 2kW. If $g = 10\ ms^{-2}$, the amount of water it can raise in one minute to a height of 10 m is :-

(1) 2000 litre

(2) 1000 litre

(3) 100 litre

(4) 1200 litre

31) A uniform chain of length and mass m overhangs a smooth table with its two third part lying on the table. Find the kinetic energy of the chain as it completely slip of the table :

(1) 0

(2) 4/9 mg

(3) 1/9 mg

(4) None of these

32) A cubic block of mass m is sliding down on an inclined plane at 60° with an acceleration of $\frac{g}{2}$, the value of coefficient of kinetic friction is

(1) $\sqrt{3}-1$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{\sqrt{2}}{3}$

(4) $1-\frac{\sqrt{3}}{2}$

33) A car is moving with uniform velocity on a rough horizontal road. Therefore, according to Newton's first law of motion

(1) No force is being applied by its engine

(2) A force is surely being applied by its engine

(3) An acceleration is being produced in the car

(4) The kinetic energy of the car is increasing

34) A block of 1 kg is stopped against a wall by applying a force F perpendicular to the wall. If μ = 0.2 then minimum value of F will be _____ N.

(1) 980

(2) 49

(3) 98

(4) 490

35) A trolley of mass 300 kg carrying a sandbag of 25 kg is moving uniformly with a speed of 27 km/h on a frictionless track. After a while, sand starts leaking out of a hole on the floor of the trolley at the rate of 0.05 kgs^{-1}. What is the speed (in km/h) of the trolley after the entire sand bag is empty ?

(1) 36

(2) 9

(3) 54

(4) 27

36) A solid disc of mass M is just held in air horizontally by throwing 20 stones per sec vertically upwards to strike the disc each with a velocity 6 ms^{-1}. If the mass of each stone is 0.05 kg what is the mass of the disc in kg (g = 10 ms^{-2}). Stones rebounds back in opposite direction with same speed after colliding with disc :

(1) 1.2

(2) 0.5

(3) 20

(4) 3

37) A block is gently placed on a conveyor belt moving horizontally with constant speed. After 4 sec. the velocity of the block becomes equal to the velocity of the belt. If the coefficient of friction between the block and the belt is $\mu = 0.2$, then the velocity of conveyor belt is -

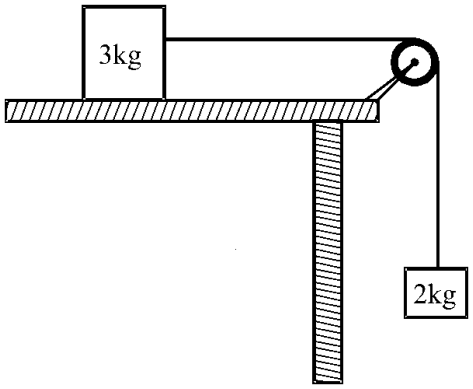
(1) 8 m/sec

(2) 4 m/sec

(3) 6 m/sec

(4) 18 m/sec

38) Consider a body of mass 3kg at rest on a smooth horizontal table. This body is connected by a light string, which passes over a smooth pulley at the edge of the table, to another body of mass 2 kg hanging freely. This 2 kg mass is released from rest. Now consider the following statements :-



(A) The masses remains at rest

(B) The 3 kg mass moves uniformly while 2 kg mass moves with acceleration $\frac{2}{5}$g m/s^2

(C) Both bodies move with acceleration $\frac{2}{5}$g m/s^2

(D) The tension in the string near the first body is more than that near the second body

(E) The tension in the string is $\frac{6g}{5}$ N

Then the correct statements are

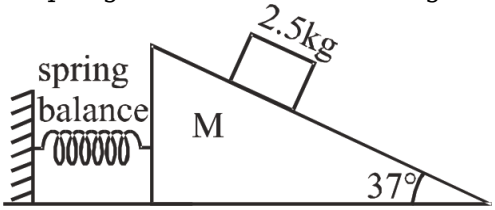
(1) B and D only

(2) C and E only

(3) B, D and E only

(4) A only

39) Find the reading of spring balance as shown in figure. Assume that mass M is in equilibrium. All surfaces are smooth.



(1) 9 N

(2) 12 N

(3) 16 N

(4) 20 N

40) A car of mass 1250 kg is moving at 30m/s. Its engine delivers 30 kW while resistive force due to surface is 750N. What max acceleration can be given in the car

(1) $\frac{1}{3}\,m/s^2$

(2) $\frac{1}{4}m/s^2$

(3) $\frac{1}{5}\,m/s^2$

(4) $\frac{1}{6}\,m/s^2$

41) A block of mass 10 kg, moving in x direction with a constant speed of 10 m/s, is subjected to a retarding force F = 0.1x J/m during its motion from x = 20 m to 30 m. Its final KE will be :

(1) 450 J

(2) 275 J

(3) 250 J

(4) 475 J

42)

A ball is thrown upwards from a planet such that distance travelled in the 4th and 5th second are equal. If the maximum height attained by the ball is 32m, choose the correct statement(s).
(A) Acceleration due to gravity of the planet is 2 m/s^2.
(B) Acceleration due to gravity of the planet is 4 m/s^2.
(C) Distance travelled by the ball in 4th second is 2m.

(D) Time taken for the ball to strike ground again is 10 s.

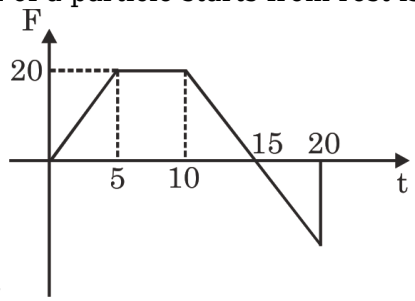
(1) A, C

(2) B, C

(3) B, C, D

(4) None of these

43) F–t, graph of a particle starts from rest is shown. At what time, linear momentum of the particle is maximum :-



(1) t = 5 sec

(2) t = 10 sec

(3) t = 15 sec

(4) t = 20 sec

44) Rate of doing work by force acting on a particle which starts its motion from origin along x-axis depends on a position x of particle and this rate is equal to 2x. Expression of velocity of particle at any position x is :-

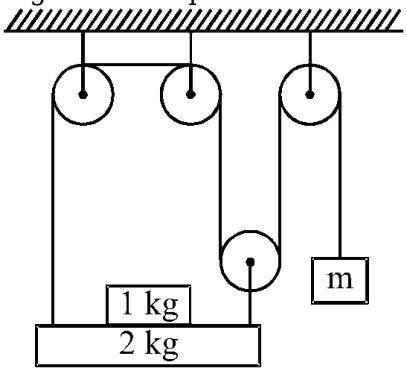
(1) $\left(\frac{3x^2}{m}\right)^{1/3}$

(2) $\left(\frac{3x^2}{2m}\right)^{1/3}$

(3) $\left(\frac{2mx}{9}\right)^{1/2}$

(4) $\left(\frac{mx^2}{3}\right)^{1/2}$

45) If the arrangement shown in the figure is in equilibrium then mass m of the block will be :-



(1) 3 kg

(2) 2 kg

(3) 1.5 kg

(4) 1 kg

CHEMISTRY

1) 12 moles of each A and B are allowed to react as given $3A + 2B \rightarrow C + \frac{1}{2} D$. If 60 gm of D is produced then, calculate the atomic mass of D :-

(1) 30
(2) 45
(3) 60
(4) 15

2) When 3g of C_2H_6 is completely burnt then find out produced volume of water vapour at STP :-

(1) 0.672 litre
(2) 6.72 litre
(3) 67.2 litre
(4) 30 litre

3) 120 g impure $CaCO_3$ sample produces 56 g of CaO on strong heating, then % purity of $CaCO_3$ sample will be :-

(1) 53.3%
(2) 83.3%
(3) 33.3%
(4) 60%

4) A sample of PCl_3 contains 1.4 mole of the substance. How many atoms are there in the sample?

(1) 4
(2) 5.6
(3) 8.431×10^{23}
(4) 3.372×10^{24}

5) What will be the weight of CO having the same number of oxygen atoms as present in 22 g of CO_2

(1) 28 g
(2) 22 g
(3) 44 g
(4) 72 g

6) In a metal oxide 32% oxygen is present. What will be equivalent mass of metal :-

(1) 34
(2) 17
(3) 32
(4) 52

7) A gaint molecule contains 0.40% of a metal by mass whose atomic weight is 39. Its minimum

molecular weight is :

(1) 19500
(2) 9750
(3) 3900
(4) 4875

8) A compound contains 38.8% C, 16% H and 45.2% N. The formula of the compound would be :-

(1) CH_3NH_2
(2) CH_3CN
(3) C_2H_5CN
(4) $CH_2(NH)_2$

9)

10^{21} molecules are removed from 200 mg of CO_2. The moles of CO_2 left are :

(1) 2.88×10^{-3}
(2) 28.2×10^{-3}
(3) 288×10^{-3}
(4) 28.8×10^{3}

10) The equivalent weight of H_3PO_4 in the given reaction is
$H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$

(1) $\frac{98}{3}$
(2) $\frac{98}{5}$
(3) $\frac{98}{2}$
(4) $\frac{98}{1}$

11) In the balanced redox reaction,
$Cr_2O_7^{2-} + SO_3^{2-} \xrightarrow{H^+} Cr^{3+} + SO_4^{2-}$
The ratio of number of Cr^{3+} ion and H^+ ion will be respectively :-

(1) 4 : 1
(2) 2 : 1
(3) 2 : 3
(4) 1 : 4

12) A sulphur-containing species that can not be a reducing agent is

(1) SO_2

(2) SO_3^{-2}

(3) H_2SO_4

(4) S^{-2}

13) Which of the following statements is not true?

(1) In a redox reaction, oxidation is always accompanied by reduction.

(2) When a negative ion changes to neutral species, the process is oxidation.

(3) Oxidising agent has a tendency to lose electrons.

(4) Conversion of MnO_4^{2-} to MnO_4^- is oxidation.

14) Equivalent weight of H_3PO_2 when it disproportionate into PH_3 and H_3PO_3 is (mol. wt. of H_3PO_2 = M) :-

(1) M

(2) $\frac{3M}{4}$

(3) $\frac{M}{2}$

(4) $\frac{M}{4}$

15) The mass of potassium dichromate required to oxidise 750 cm^3 of 0.6 M Mohr's salt solution is :- [M. wt. $K_2Cr_2O_7$ = 294]

(1) 0.49 g

(2) 0.45 g

(3) 22.05 g

(4) 2.2 g

16) In the half rection, the number of electrons needed to balance the rection would be-
$N_2H_4 \rightarrow 2NO_3^-$

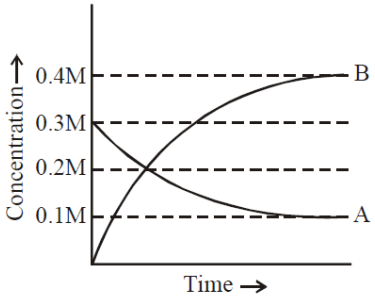
(1) 10 e^-

(2) 8 e^-

(3) 5 e^-

(4) 14 e^-

17) The figure shows the change in concentration of species A and B as a function of time. The



equilibrium constant K_c for the reaction A(g) ⇌ 2B(g) is :-

(1) $K_c > 1$

(2) $K < 1$

(3) $K = 1$

(4) Data insufficient

18)

For the reaction $A(g) \rightleftharpoons 2B(g)$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500, at 1000 K. [**Given :** $R = 0.0831$ L atm mol^{-1} K^{-1}]
K_P for the reaction at 1000 K is

(1) 83.1

(2) 2.077×10^5

(3) 0.033

(4) 0.021

19) 2 mole of PCl_5 were introduced in a vessel of 5L capacity at a particular temperature. At equilibrium 35% of PCl_5 is dissociated into PCl_3 and Cl_2. The total number of moles at equilibrium is :-

(1) 1.89

(2) 3.7

(3) 1.33

(4) 2.7

20) **Assertion :-** For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$ on increasing temperature and increasing pressure more water will form.
Reason :- Since forward reaction is endothermic in nature and volume of water is greater than that of the volume of ice.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

21) The equilibrium constant for reaction
$2A + B \rightleftharpoons 2C$ is 40,
then equilibrium constant for reaction $C \rightleftharpoons A + \frac{1}{2} B$:

(1) $\frac{1}{40}$

(2) 40

(3) $\sqrt{\frac{1}{40}}$

(4) $\left(\frac{1}{40}\right)^2$

22)

For the reaction $2A \rightleftharpoons B + C$, $K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is:

$[A] = [B] = [C] = 2 \times 10^{-3}$ M.
Then, which of the following is correct ?

(1) Reaction is at equilibrium.
(2) Reaction has a tendency to go in forward direction.
(3) Reaction has a tendency to go in backward direction.
(4) Reaction has gone to completion in forward direction.

23) At equilibrium PCl_5 is dissociating 50% at 25°C at a total pressure of P atm. If equilibrium constant is K_P, then which of the following relation is numerically correct ?

(1) $K_P = 3P$
(2) $P = 3K_P$
(3) $P = 2K_P/3$
(4) $K_P = 2P/3$

24) Which of the following statement is incorrect ?

(1) On addition of catalyst the equilibrium constant value is not affected.
(2) Equilibrium constant for a reaction with negative ΔH value decreases as the temperature increases.
(3) In equilibrium mixture of ice and water kept in perfectly insulated flask, mass of ice and water does not change with time
(4) All the measurable properties of the system are variable at equilibrium state

25) In the reaction $2A(g) + B(g) \rightleftharpoons C(g) + 242$ kcal, which one of the following conditions will favour maximum formation of the product :

(1) Low temperature and high pressure
(2) Low temperature and low pressure
(3) High temperature and high pressure
(4) High temperature and low pressure

26) Which of the following is correct relation for the solution of the salt CH_3COONH_4

Concentration	Degree of hydrolysis
0.001 M	x_1
0.002 M	x_2

(1) $x_1 = x_2$
(2) $x_1 > x_2$
(3) $x_1 < x_2$

(4) $x_1 >> x_2$

27) What volume of 0.2 M $Ba(OH)_2$ solution should be mixed with 5L, 0.1 M H_2SO_4 solution to get a solution which have no effect on litmus paper :-

(1) 1.5 L
(2) 2 L
(3) 2.5 L
(4) 3 L

28) In the following which one has highest degree of hydrolysis :

(1) 0.01 M NH_4Cl
(2) 0.1 M NH_4Cl
(3) 0.001 M NH_4Cl
(4) Same in all

29) The solution of Na_2CO_3 is alkaline due to :

(1) hydrolysis of CO_3^{2-} ions
(2) hydrolysis of Na^+ ions
(3) neutralisation of CO_3^{2-} ions
(4) neutralisation of Na^+ ions

30) An aqueous solution at room temperature contains 0.1 M NH_4Cl and 0.01 M $NH_4OH(pK_b = 5)$ pH of this solution is :

(1) 7.5
(2) 6.8
(3) 6.5
(4) 8.0

31) Solution A has 1 mol each of NaOH and NH_4Cl per litre and solution B has 1 mol each of NH_4OH and NH_4Cl per litre. What will be the ratio of pOH value of solution A and B.

(1) 1 : 2
(2) 1 : 1
(3) 1 : 3
(4) 1 : 4

32) Find the minimum pH required to precipitate $Pb(OH)_2$ ($K_{sp} = 1.2 \times 10^{-5}$) from 0.12 M $PbCl_2$ solution.

(1) 12.4
(2) 10.8

(3) 12

(4) 11.1

33) The entropy values (in J K^{-1} $mo1^{-1}$) of $H_2(g)$ = 130.6, $Cl_2(g)$ = 223 and HCl(g) = 186.7 at 298 K and 1 atm pressure are given. Then entropy change for the reaction,
$H_2(g) + Cl_2(g) \rightarrow 2HCl$

(1) +540.3

(2) +727.3

(3) –166.9

(4) +19.8

34) Match the Column-I to Column-II.

	Column-I (Sign of ΔH & ΔS respectively)		**Column-II (Nature of reaction)**
(A)	– & –	(P)	Spontaneous only at low temperature
(B)	– & +	(Q)	Spontaneous only at high temperature
(C)	+ & +	(R)	Spontaneous at all temperature
(D)	+ & –	(S)	Nonspontaneous at all temperature

(1) (A)–(P); (B)–(R); (C)–(Q); (D)–(S)

(2) (A)–(R); (B)–(P); (C)–(S); (D)–(Q)

(3) (A)–(S); (B)–(Q); (C)–(R); (D)–(P)

(4) (A)–(Q); (B)–(P); (C)–(S); (D)–(R)

35) At 27°C one mole gas is allowed to expand reversibly and adiabatically from 10 L to 20 L. Then calculate change in entropy :-

(1) Zero

(2) + 2.5 cal/K

(3) 1.38 cal/K

(4) 5.72 cal/K

36) A piston filled with 0.04 mol of an ideal gas expand reversibily from 50.0 mL to 375 mL at a constant temperature of 37ºC. As it does so it absorb 208 J of heat. The value of q and w for the process will be :-
(R = 8.314 J / mol K) (ln 7.5 =2.01)

(1) q = – 208 J, w = + 208 J

(2) q = + 208 J, w = + 208 J

(3) q = + 208 J, w = – 208 J

(4) q = – 208 J, w = – 208 J

37) At 298 K in Bomb calorimeter, heat of combustion of methane gas is –800 kJ/mol. Calculate its

enthalpy of combustion at constant pressure at same temperature ?

(1) –805 kJ

(2) 805 kJ

(3) –795 kJ

(4) 795 kJ

38) Enthalpy of combustion of carbon to carbon dioxide is –390.0 kJ mol^{-1}. The amount of heat released, when 35.0 g of CO_2 is formed from the reaction of carbon and dioxygen gas, is :

(1) 490 kJ

(2) 245 kJ

(3) 700 kJ

(4) 310 kJ

39) The standard enthalpy of neutralization is –57.3 kJ mol^{-1} as :

$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(\ell)}$,

strong acid strong base

ΔH = –57.3 kJ mol^{-1}

The enthalpy of neutralization of 0.25 mole of HCl by 0.25 mol of NaOH is :

(1) –28.3 kJ

(2) –14.32 kJ

(3) +57.3 kJ

(4) –57.3 kJ

40) Calorific value of hydrogen gas is –143 kJg^{-1}. The standard enthalpy of formation of H_2O will be -

(1) – 143 kJ mol^{-1}

(2) + 143 kJ mol^{-1}

(3) – 286 kJ mol^{-1}

(4) + 286 kJ mol^{-1}

41) Calculate ΔH for the given reaction :

$2C(s) + H_2(g) \rightarrow C_2H_2(g)$

Given -

$C_2H_2(g) + \frac{5}{2}O_2(g) \rightarrow 2CO_2(g) + H_2O(\ell)$;

ΔH = – 1299.5 kJ

$C(s)+O_2(g) \rightarrow CO_2(g)$; ΔH = – 393.5 kJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$; ΔH = – 285.8 kJ

(1) +226.7 kJ

(2) –226.7 kJ

(3) –619.7 kJ

(4) +619.7 kJ

42) $N_2(g) + 2O_{2(g)} \rightarrow 2NO_2$; $\Delta H = + x$ kJ
$2NO(g) + O_{2(g)} \rightarrow 2NO_2(g)$; $\Delta H = + y$ kJ
The enthalpy of formation of NO is -

(1) (2x – 2y) kJ/mol

(2) (x – y) kJ/mol

(3) $\frac{1}{2}$ (y – x) kJ/mol

(4) $\frac{1}{2}$ (x – y) kJ/mol

43) The bond enthalpies of $H_{2(g)}$, $X_{2(g)}$ and $HX_{(g)}$ are in the ratio of 2 : 1 : 2. If the enthalpy for formation of $HX_{(g)}$ is –50 kJ mol^{-1}, the bond enthalpy of H_2 is

(1) 200 kJ mol^{-1}

(2) 400 kJ mol^{-1}

(3) 100 kJ mol^{-1}

(4) 300 kJ mol^{-1}

44) The standard enthalpy of formation of C_7H_{16} is –200 kJ/mol. The enthalpy of formation of CO_2 and H_2O are – 394 kJ/mol and –286 kJ/mol respectively. The enthalpy of combustion of C_7H_{16} is :-

(1) –5476 kJ/mol

(2) –4846 kJ/mol

(3) –5200 kJ/mol

(4) –5726 kJ/mol

45)

The enthalpy of the reaction,
$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)} + Q_1$ and that of is
$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(\ell)} + Q_2$ then

(1) $Q_1 > Q_2$

(2) $Q_1 + Q_2 = 0$

(3) $Q_1 < Q_2$

(4) $Q_1 = Q_2$

BIOLOGY

1) Choose correct option for A, B and C ?
Solanum + A = Solanaceae
Panthera + B = C

	A	B	C
(1)	*Petunia*	*Canis*	Canidae
(2)	*Datura*	*Felis*	Felidae
(3)	*Mangifera*	*Canis*	Canidae
(4)	*Oryza*	*Felis*	Felidae

(1) 1

(2) 2

(3) 3

(4) 4

2) Which of the following would not lead to increase in number of organisms ?
(A) True regeneration
(B) Binary fission
(C) Regeneration
(D) Endospore formation

(1) A, C & D

(2) A, B & C

(3) C & D

(4) A & D

3) Study the following statements carefully and give the correct answer :-
(A) Lower the taxa, more are the characteristics that the members within the taxon share.
(B) Higher the taxonomic category, greater is the difficulty of determining the relationships to other taxa at the same level.
(C) In lower taxa the problems of classification becomes more complex.
(D) As we go higher from species to kingdom, the number of common characters increases.

(1) A,B – incorrect ; C, D – correct

(2) A,B – correct ; C,D – incorrect

(3) B, D – correct ; A, C – incorrect

(4) A,C – correct ; B, D – incorrect

4) Consider the followings- *Solanum tuberosum, Mangifera indica, Triticum aestivum, Solanum nigrum, Panthera leo, Panthera pardus, Solanum melongena, Panthera tigris, Musca domestica* How many family, genus and species are represented by above set of organisms?

(1) 5, 5, 5 respectively

(2) 5, 5, 9 respectively

(3) 9, 6, 5 respectively

(4) 6, 5, 9 respectively

5) **Statement-I :** Numerical taxonomy uses computers.
Statement-II : Phylogenetic classification is not based on evolutionary relationships.

(1) Statement I and II are correct

(2) Statement I & II are incorrect

(3) Statement I is correct & II is incorrect

(4) Statement II is correct & I is incorrect

6) Read the following statements carefully & find out correct statements for five kingdom system:
(A) Three kingdoms contain autotrophic organisms
(B) Only two kingdoms are of multicellular organisms
(C) Four kingdoms contain nuclear membrane
(D) All kingdoms contain chemosynthetic autotrophic organism

(1) C & D

(2) A & B

(3) A & C

(4) B & D

7) Read the following statements and choose the correct option.
Statement-1 : Two kingdom classification system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organsims.
Statement-2 : Cell wall of bacteria is made up of polysaccharide and amino acid and in fungi it is made up of chitin.

(1) Both statement-1 and 2 are correct

(2) Both statement-1 and 2 are incorrect

(3) Only statement 1 is correct

(4) Only statement 2 is correct

8) Read the following statements carefully :-
(A) Some bacteria have resistence to certain antibiotics, this ability is provided by the extra genomic DNA of bacteria
(B) Mesosomes help in respiration and DNA replication
(C) Pili and fimbriae are the surface appendages of bacterial cell which play a role in motility.
(D) Glycocalyx, cell wall and cell membrane form the cell envelope of bacterial cell.
Which of the above statement(s) is/are correct and incorrect ?

	Correct statement	**Incorrect Statement**
(1)	A,B & D	C
(2)	A & B	C & D
(3)	B & D	A & C
(4)	B, C & D	A

(1) 1

(2) 2

(3) 3

(4) 4

9) During a microbial study, a spore was observed which was thick walled, highly resistant and surrounded by different layers. These can withstand temperature as high as 100°C or as low as –100°C, so they can remain unharmed during pasteurisation. From these characters spore must be :-

(1) Zoospores

(2) Endospore

(3) Auxospore

(4) Conidia

10) What is incorrect for chemosynthetic autotrophic bacteria :-

(1) Oxidise various organic and inorganic substances

(2) Use the energy released through oxidation for their ATP production

(3) Play great role in recycling nutrients like nitrogen, phosphorus, iron and sulphur

(4) They are most abundant in nature

11) Consider the following four statements (A–D) and select the option which includes all the correct ones only.
(A) Archaebacteria differ from other bacteria in having a different cell wall structure
(B) Bacteria as a group show the most extensive metabolic diversity
(C) Cyanobacteria have chlorophyll 'b' similar to green plants.
(D) *Mycoplasma* can not survive without oxygen.

(1) Statements B, C and D only

(2) Statements A and C only

(3) Statements B and D only

(4) Statements A and B only

12) Match Column-I with Column-II and select the correct option from the given codes :-

	Column-I		**Column-II**
A	Red sea	i	Euglenoids
B	Red tides	ii	*Trichodesmium erythraeum*
C	Mixotrophic nutrition	iii	Slime moulds
D	Plasmodium	iv	Dinoflagellates

(1) A-(ii),B-(iv), C-(i), D-(iii)

(2) A-(ii),B-(iv), C-(iii), D-(i)

(3) A-(ii),B-(iii), C-(i), D-(iv)

(4) A-(i),B-(iv), C-(iii), D-(ii)

13)

Read the following pair :-
(A) Diatoms-Chief producer in Ocean

(B) Dinoflagellates-Thin proteinous cell wall
(C) Slime mould-Spores are resistant under adverse conditions.
(D) Euglenoids-Cellulosic cell wall

(E) Protozoans-Unicellular eukaryotes
Choose the option in which all the statements are correct :-

(1) A,B,C only

(2) C,D,E only

(3) A,C,E only

(4) A,B,E only

14) How many of the following statements are correct ?
(A) Protists reproduce asexually and sexually and some have flagella or cilia.
(B) Slime moulds are saprophytic protists.
(C) The cell wall of dinoflagellates has stiff silicated plates on outer surface.
(D) Diatoms float passively with water currents
(E) The pigments of euglenoids are identical to those present in higher plants.

(1) Five

(2) Four

(3) Three

(4) Two

15) Match the column I with column II and choose the correct option.

	Column I		**Column II**
(i)	*Euglena*	(a)	*Physarum*
(ii)	Diatom	(b)	Diatomaceous earth
(iii)	Slime mould	(c)	Biflagellated
(iv)	Dinoflagellates	(d)	*Gonyaulax*
		(e)	Pellicle
		(f)	Chrysophytes

(1) (i)-e, (ii)-b,f (iii)-c, (iv)-a,d

(2) (i)-e, (ii)-b,f (iii)-a, (iv)-c,d

(3) (i)-e, (ii)-b,c,f (iii)-a, (iv)-d

(4) (i)-e, (ii)-b,c,f (iii)-d, (iv)-a

16) Select the incorrect match :-

(1) Morels and truffles – Phycomycetes

(2) Puff balls and toad stools – Basidiomycetes

(3) Deuteromycetes – *Alternaria solani*

(4) Late blight of potato – *Phytopthora infestans*

17) After karyogamy followed by meiosis, spores are produced exogenously in :-

(1) *Trichoderma*
(2) *Saccharomyces*
(3) *Neurospora*
(4) Rust fungi

18) Select the incorrect match pair from the following:-

(1) Algal fungi – Presence of long intervening dikaryotic stage
(2) Sac fungi – Sexual spores are ascospores
(3) Club fungi – Sex organs are absent
(4) Imperfect fungi – Presence of exogeneous asexual conidia.

19) Find out the incorrect statement–

(1) Most of the fungi are hetrotrophic.
(2) Fungi prefer to grow in warm and humid places.
(3) Wheat rust is caused by *Albugo.*
(4) Yeast is used in making bread and beer.

20) **Statement 1 :** Fungi are cosmopolitan which occur in air, water, soil as well as on animals and plants.
Statement 2 : Fungi show a great diversity in morphology and habitat.

(1) Both Statement 1 & Statement 2 are true.
(2) Statement 1 is false but Statement 2 is true.
(3) Statement 1 is true but the Statement 2 is false.
(4) Both Statement 1 & Statement 2 are false.

21) I. ___A___ commonly known as imperfect fungi because only the asexual or vegetative phase of these fungi are known.
II. The mycelium of bracket fungi is ____B____ and ___C____.
III. The asexual spore of sac fungi are conidia produced ____D_____ on the special mycelium.
Identify the A, B, C and D -

	A	B	C	D
1.	Ascomycetes	Septate	Unbranched	Exogenously
2.	Deuteromycetes	Septate	Branched	Exogenously
3.	Basidiomycetes	Aseptate	Unbranched	Endogenously
4.	Deuteromycetes	Septate	Unbranched	Endogenously

(1) 1
(2) 2
(3) 3

(4) 4

22) Read the following statements :-
(A) Lichens do not grow in polluted areas.
(B) Viroids have low molecular weight DNA
(C) Virus are inert outside their specific host cell.
(D) AIDS, small pox and cholera disease caused by Bacteria.
Which statements are correct ?

(1) A and B

(2) A and C

(3) A, B and C

(4) A, B and D

23) Read the following statements carefully
(A) Viruses are smaller than bacteria
(B) Viruses are always made up of DNA and protein only
(C) Viruses contain approximately 5% RNA and 95% DNA
(D) Viruses contain nucleic acid so they are capable of protein synthesis independently.
Which of the above statements is/are correct ?

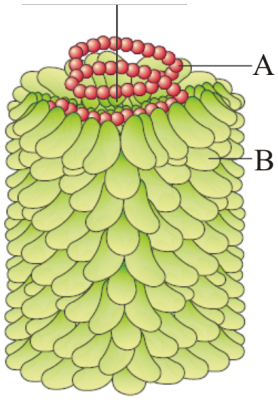
(1) A and B

(2) A, B and D

(3) B and C

(4) A only

24) Study the given figure for structure of TMV (Tobacco Mosaic Virus) and select the option that



correctly identifies the labellings A and B.

	A	B
(1)	ssDNA	Capsomeres
(2)	dsDNA	Capsomeres
(3)	ssRNA	Capsomeres
(4)	dsRNA	Tail fibres

(1) 1

(2) 2

(3) 3

(4) 4

25) **Assertion :** In lichens, mycobiont and phycobiont are symbiotically associated in which algae is predominant and fungi is a subordinate partner.
Reason : The fungus provides food and algae protects the fungus from unfavourable conditions.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

26) Match Column I with Column II and select the correct option from the codes given below–

Column I		**Column II**	
(A)	Monera	(i)	*Spirogyra* and *Solanum*
(B)	Protista	(ii)	*Bacillus, Oscillatoria*
(C)	Fungi	(iii)	*Euglena, Trypanosoma*
(D)	Plantae	(iv)	*Mucor, Penicillium*
(E)	Animals	(v)	*Felis, Panthera*

(1) A–(iii), (B)–(ii), (C)–(iv), (D)–(i), (E)–(v)
(2) A–(ii), (B)–(iii), (C)–(iv), (D)–(i), (E)–(v)
(3) A–(ii), (B)–(iii), (C)–(i), (D)–(iv), (E)–(v)
(4) A–(ii), (B)–(v), (C)–(i), (D)–(iv), (E)–(iii)

27) Match the List-I with List-II and select the correct option –

List-I		**List-II**	
A	Viroids	(i)	Production of biogas
B	Methanogens	(ii)	Parasitic fungi on mustard
C	*Albugo*	(iii)	Cr-Jacob disease
D	Prions	(iv)	T.O. Diener

(1) A-(i), B-(ii), C-(iv), D-(iii)
(2) A-(iv), B-(iii), C-(i), D-(ii)
(3) A-(iv), B-(i), C-(ii), D-(iii)
(4) A-(ii), B-(i), C-(iii), D-(iv)

28) Match the column I with column II :-

Particles/Micro organisms		**Diseases**	
(i)	Viroid	(a)	Abnormally folded protein

(ii)	Virus	(b)	Citrus canker
(iii)	Bacteria	(c)	Potato spindle tuber disease
(iv)	Prion	(d)	Mumps

Options are :-

(1) i - a, ii - b, iii - d, iv - c

(2) i - c, ii - d, iii - a, iv - b

(3) i - c, ii - d, iii - b, iv - a

(4) i - a, ii - c, iii - b, iv - d

29) Which one of the following options, gives the correct categorisation of six organism to the type of stored food (A, B, C) they have.

	A Floridean Starch	**B Starch**	**C Laminarian**
(1)	*Porphyra Fucus*	*Chlamydomonas Volvox*	*Laminaria Chara*
(2)	*Fucus Gelidium*	*Spirogyra Dictyota*	*Porphyra Gracilaria*
(3)	*Porphyra Gracilaria*	*Chara Dictyota*	*Sargassum Polysiphonia*
(4)	*Polysiphonia Porphyra*	*Volvox Ulothrix*	*Ectocarpus Laminaria*

(1) 1

(2) 2

(3) 3

(4) 4

30) Consider the genera given below and answer the question asked next to them :
Gelidium, Chlorella, Chara, Fucus, Dictyota, Sargassam, Volvox, Gracilaria.
How many belongs to Pheophyceae and Rhodophyceae :-

(1) Three and two

(2) Five and four

(3) Four and three

(4) Four and five

31) Oogamous type sexual reproduction are found in -

(1) *Volvox, Fucus*

(2) *Volvox, Eudorina*

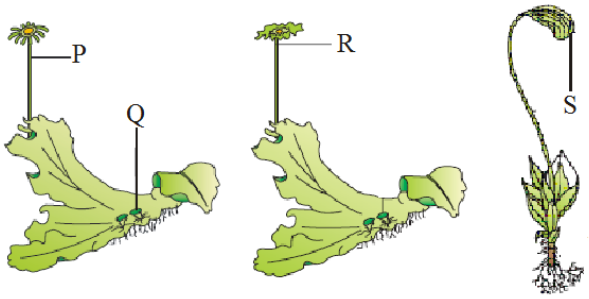
(3) *Fucus, Ulothrix*

(4) *Albugo, Spirogyra*

32) Which of the following is true for alternation of generation :–

(1) The sporophyte undergoes syngamy to produce spores
(2) The sporophyte undergoes meiosis to produce spores
(3) The gametophyte undergoes syngamy to produce spores
(4) The gametophyte undergoes meiosis to produce gametes

33) Observe the following figures and identify the labelled structures P, Q, R and S :-



	P	Q	R	S
1.	Archegoniophore	Gemma cup	Antheridiophore	Seta
2.	Antheridiophores	Rhiziods	Archegoniophore	Capsule
3.	Antheridiophore	Seta	Archegoniophore	Leaves
4.	Archegoniophore	Gemma cup	Antheridiophore	Capsule

(1) 1
(2) 2
(3) 3
(4) 4

34) Consider the following statements about **bryophytes**.
I. Sexual reproduction is oogamous type.
II. The sex organs are multicellular and surrounded with sterile jacket.
III. The haploid gametophyte is dominant stage in the life cycle of bryophytes.
Which of the statements given above are correct?

(1) I and II only
(2) I and III only
(3) II and III only
(4) I, II and III

35) The correct sequence of events in the life cycle of bryophytes is :-
A : Production of antherozoids and egg
B : Fusion of gametes
C : Formation of sporophyte
D : Meiosis in spore mother cell
E : Germination of spores

(1) A, D, B, C, E

(2) A, B, C, E, D

(3) A, B, C, D, E

(4) A, E, D, C, B

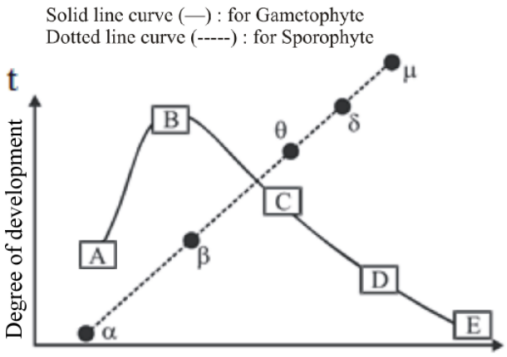
36) **Assertion :** Heterosporous condition in pteridophytes is considered as an event which is precursor to seed habit.
Reason : The female gametophytes in heterosporous pteridophytes are retained on the parent sporophytes for variable periods and the development of the zygotes into young embryo take place within the female gametophytes.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

37) 

In given graph the curves are showing the evolutionary changes in gametophyte and sporophyte of different types of plants. Which of the following point represents the last group of gametophytic generations showing antheredia ?

(1) A

(2) B

(3) C

(4) D

38) Homosporous fern plants produce :-

(1) Two types of spores

(2) Dioecious gametophytes

(3) Monoecious gametophytes

(4) Leafy gametophytes

39) **Assertion (A) :** In gymnosperms the male and female gametophytes do not have an independent free living existence.
Reason (R) : In gymnosperm, spores are germinated in soil.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

40) **Statement-I :** In Gymnosperm, microspores develop into a male gametophytic generation and is confined to a limited number of cells.
Statement-II : In Gymnosperm, microspore develop into pollen grain or male gametophyte within microsporangia.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect.
(3) Only Statement-I is correct.
(4) Only Statement-II is correct.

41) Given below are two statements :-
Statement-I : In gymnosperms, one of the megaspore develops into multicellular structure called female gametophyte that bears two or more archegonia & the multicellular female gametophyte is retained within megasporangium.
Statement-II : Unlike the gymnosperms where the ovules are naked, in the angiosperms, the pollen grains and ovules are developed in specialised structures called flowers.
In the light of the above statements. Choose the most appropriate answer from the option given below :

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are incorrect.
(4) Both Statement-I and Statement-II are correct.

42) Read the following examples.
Lycopodium, Selaginella, Cycas, Pinus, Psilotam, Eucalyptus, Funaria, Salvinia, Sphagnum, Chara
How many examples have heterosporous nature?

(1) Three
(2) Four
(3) Five
(4) Six

43)

Green algae usually have a rigid cell wall made of :

(1) An inner layer of pectose and outer layer of cellulose
(2) An inner layer of cellulose and outer layer of pectose
(3) An inner layer of protein and outer layer of pectose
(4) An inner layer of cellulose and outer layer of protein

44) Which of the following plant have coralloid roots?

(1) *Gnetum*
(2) *Cedrus*
(3) *Cycas*

(4) *Pinus*

45) The given table shows classes of algae and their main characteristics. Some characteristics in this table mentioned as A, B, C and D, Observe the table and choose the option in which A, B, C and D are correctly identified :-

Classes	Chlorophyceae	Phaeophyceae	Rhodophyceae
Major Pigments	Chl a, b	Chl a, c	A
Stored Food	Starch	B	Floridean starch
Cell Wall	C	Cellulose, algin	Cellulose, pectin, polysulphate ester
Flagellar Number	2-8	2	D

	A	B	C	D
(1)	Chl a, e	Fat & oil	Peptidoglycan	Many
(2)	Chl a, d	Mannitol, Laminarin	Cellulose	Absent
(3)	Chl a, d	Paramylum	Lipoprotein	One
(4)	Chl a, e	Glycogen	Cellulose	Absent

(1) 1

(2) 2

(3) 3

(4) 4

46) Given below are two statements :
One is labeled as Assertion (A) and the other is labeled as Reason (R) :
Assertion (A) : Animals like Nereis and Earthworm are placed under phylum Annelida.
Reason (R) : Body surface of these animals (Nereis, earthworm) is distinctly marked out into segments or metameres.

(1) Both Assertion and Reason but Reason is not the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason true and Reason is the correct explanation of Assertion.

47) Which of the following is the incorrect statement regarding phylum - Echinodermata ?

(1) All are marine with organ system level of organisation.

(2) Digestive system is complete with mouth on the lower (ventral) and anus on the upper (dorsal) side.

(3) An excretory system is absent.

(4) Sexes are not separate.

48) Which of the following is the correct match of animal with it's common name ?

(1) Ophiura - Brittle star
(2) Antedon - Sea urchin
(3) Asterias - Sea-lily
(4) Echinus - Star fish

49) Which one is not the charater of Osteichthyes?

(1) Terminal mouth
(2) Gills with operculum
(3) Body covered by placoid scales
(4) Air bladder regulates buoyancy.

50) How many organisms are releated with Echinodermata phylum ?
Star fish, Sea-urchin, Sea-lily, Sea-hare,
Sea-cucumber, Brittle star, Sea-fan, Sea-pen

(1) 4
(2) 5
(3) 6
(4) 2

51) Bath sponge is common name of -

(1) Spongilla
(2) Euspongia
(3) Euplectella
(4) Leucosolenia

52)

Consider the following statements :-
(A) Protochordates are exclusively marine
(B) In cephalopods, notochord extends from head to tail region
(C) In urochordates, notochord is present only in larval tail
(D) Cranium and vertebral column are bony in cyclostomates

Which of the above statement is/are correct ?
(1) A and C
(2) B, C and D
(3) A, B and C
(4) All of these

53) Select the incorrect option for the following given figure.



(1) Circulation closed type

(2) Adult migrate from sea to river

(3) Scales and paired fins present

(4) Ectoparasite

54) In which of the following operculum and air bladder both are present, sexes are separate, fertilisation is usually external and male does not bear claspers as copulatory organ ?

(1) Torpedo

(2) Labeo

(3) Pristis

(4) Trygon

55) Match the following columns and select the correct option :-

Column-I		**Column-II**	
(A)	Ctenophora	(I)	*Sycon*
(B)	Annelida	(II)	*Pheretima*
(C)	Mollusca	(III)	*Bombyx*
(D)	Echinodermata	(IV)	*Locusta*
		(V)	*Pinctada*
		(VI)	*Echinus*
		(VII)	*Saccoglossus*
		(VIII)	*Ctenoplana*

(1) (A)-(VIII), (B)-(V), (C)-(VI), (D)-(VII)

(2) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(3) (A)-(V), (B)-(VI), (C)-(VII), (D)-(VIII)

(4) (A)-(VIII), (B)-(II), (C)-(V), (D)-(VI)

56) **Assertion :** Cartilage and Bones are specialised connective tissue.
Reason : Cartilage is solid pliable and resists compression while bone have a hard and non-pliable ground substance rich in calcium and collagen fibres.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

57) **Statement-I :** Digestive system of echinoderms is complete with mouth on the lower (Ventral)

side and anus on the upper (dorsal) side.
Statement-II : The most distinctive feature of echinoderms is the presence of water transport system or canal system which helps in locomotion, capture and transport of food and respiration.

(1) Statement-I and II both are incorrect
(2) Statement-I and II both are correct
(3) Only statement-I is correct
(4) Only statement-II is correct

58) Read the following statements carefully and answer the question given below.
(a) Cyclostometas have circular and sucking mouth without jaw.
(b) Chondrichthyes have a gill slits without operculumn.
(c) Osteichthyes have 5-7 pairs of gills covered by operculum.
(d) In Amphibia alimentary canal, ureter & reproductive tract open in anus.
How many of the following options is/are incorrect.

(1) (b) & (c)
(2) (c) & (d)
(3) Only (d)
(4) (b) & (d)

59) In which animal body is divided into head, muscular foot and visceral hump ?

(1) *Ascaris*
(2) Earthworm
(3) Cockroach
(4) *Pila*

60) The characteristic feature of class Aves is

(1) Presence of tail
(2) Moist skin
(3) Warm blooded
(4) Presence of feathers

61) In which of the following chordate, notochord is absent in adult form?

(1) *Scoliodon*
(2) *Salpa*
(3) *Lancelet*
(4) All of the above

62) Match the column & select correct option :

	Column-I		**Column-II**
(i)	Dog fish	(a)	Scoliodon

(ii)	Magur	(b)	Clarias
(iii)	Flying fish	(c)	Pristis
(iv)	Saw fish	(d)	Exocoetus

(1) (i)-a, (ii)-b, (iii)-c, (iv)-d

(2) (i)-b, (ii)-a, (iii)-c, (iv)-d

(3) (i)-d, (ii)-a, (iii)-c, (iv)-b

(4) (i)-a, (ii)-b, (iii)-d, (iv)-c

63) **Statement-I :-** In platyhelminthes the mesoderm is present as scattered, pouches, between ectoderm and endoderm.
Statement-II :- In earthworms, the body is externally and internally divided into segments with a serial repetition of some organs.

(1) Statement I is correct but II is incorrect.

(2) Statement II is correct but I is incorrect.

(3) Both statements are correct.

(4) Both statements are incorrect.

64) _____ forms a diffusion boundary in our body

(1) Compound epithelium

(2) Simple columnar epithelium

(3) Simple squamous epithelium

(4) Simple cuboidal epithelium

65) Cuboidal epithelial cells with microvilli are present in :-

(1) PCT of Nephron

(2) Bowman's capsule

(3) Thin part of Henle's loop

(4) Alveoli of lungs

66) Match the following columns :-

Column-I		**Column-II**	
A	Areolar tissue	1	Fat cells
B	Adipose tissue	2	Osteocytes
C	Ligaments	3	Loose connective tissue
D	Bone	4	Dense regular connective tissue

(1) A – 3, B – 1, C – 4, D – 2

(2) A – 1, B – 2, C – 3, D – 4

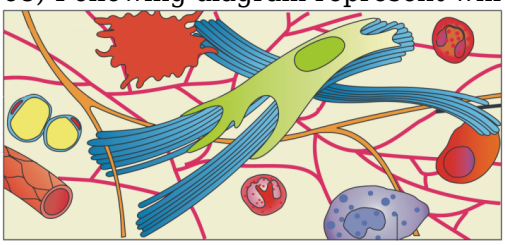
(3) A – 4, B – 3, C – 2, D – 1

(4) A – 2, B – 1, C – 4, D – 3

67) The type of epithelium which line the inner surface of fallopian tubes -

(1) Squamous epithelium
(2) Compound epithelium
(3) Ciliated epithelium
(4) Cuboidal epithelium

68) Following diagram represent which type of connective tissue ?



(1) Specialized connective tissue
(2) Dense connective tissue
(3) Adipose tissue
(4) Areolar connective tissue

69) **Statement I :** Some of the columnar or cuboidal cells get specialised for secretion and are called glandular epithelium.
Statement II : Compound epithelium is made up of single layer of cells and thus has limited role in secretion and absorption.

(1) Statement I and II both are correct.
(2) Statement I and II both are incorrect.
(3) Only statement I is correct.
(4) Only statement II is correct.

70) Match the following

	Column-A		**Column-B**
P	Tight junction	i	rapid transfer of ions
Q	Gap junctions	ii	limited role in secretion and absorption
R	Compound epithelium	iii	bronchioles
S	Microvilli	iv	PCT
		v	stop substances from leaking across a tissue

(1) P-(i), Q-(ii), R-(iii), S-(iv)
(2) P-(ii), Q-(i), R-(iii), S-(iv)
(3) P-(i), Q-(v), R-(iii), S-(ii)
(4) P-(v), Q-(i), R-(ii), S-(iv)

71) The sclerite present on ventral side of cockroach is known as :

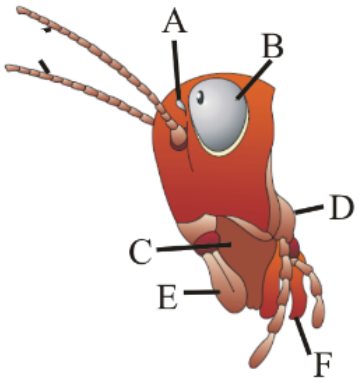
(1) Tergites

(2) Pleurites

(3) Sternites

(4) Tergum

72) The figure given below shows the head region of cockroach. Identify A to F.



(1) A-Compound eye, B-Ocellus, C-Maxilla, D-Mandible, E-Labrum, F-Labium

(2) A-Ocellus, B-Compounds eye, C-Mandible, D-Maxilla, E-Labrum, F-Labium

(3) A-Ocellus, B-Compound eye, C-Mandible, D-Maxilla, E-Labium, F-Labrum

(4) A-Ocellus, B-Compound eye, C-Maxilla, D-Mandible, E-Labrum, F-Labium

73) Which of the following statements is/are incorrect about Periplaneta americana ?
A. They are nocturnal omnivores that live in the damp places.
B. Its body is segmented and divisible in two region-head and abdomen
C. Antennae have sensory receptor to monitor the environment.
D. Head can move in all directions due to the presence of movable neck.
The correct option is -

(1) A and D

(2) only B

(3) only D

(4) B and C

74) Which one of the following statement is correct for cockroach ?

(1) The number of ovarioles in each ovary are ten

(2) The larva stage is called caterpillar

(3) Anal styles are absent in females

(4) They are ureotelic

75) Collaterial glands are found in :

(1) Female cockroach

(2) Male cockroach

(3) Both 1 and 2

(4) Frog

76) Which of the following excretory structure absent in female cockroach ?

(1) Malpighian tubule
(2) Urecose gland
(3) Fat body
(4) Nephrocyte cell

77) In cockroach, a ring of 6-8 hepatic or gastric caeca is present :-

(1) In gizzard
(2) In ileum
(3) At the junction of foregut and mid gut
(4) At the junction of mid gut and hind gut

78) Given below are two statements :
Statement-I :- The head of the cockroach is connected with thorax by a short extension of the head known as the neck.
Statement-II :- When resting, the hind wings of the cockroach are covered by the fore wings.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both statement-I and statement-II are false.
(2) Statement-I is true but statement-II is false.
(3) Statement-I is false but statement-II is true.
(4) Both statement-I and statement-II are true.

79) The number of nerve ganglia in the thorax and abdomen of cockroach are respectively :

(1) Three and seven
(2) Four and seven
(3) Three and nine
(4) Three and six

80) Sinus venosus in the heart of frog joins the

(1) Left atrium
(2) Ventricle
(3) Truncus arteriosus
(4) Right atrium

81) Frogs exhibit sexual dimorphism due to presence of -

(1) Tympanum
(2) Vocal sac
(3) Copulatory pad

(4) 2 & 3 both

82) Which organ is not functionally connected with kidney in female reproductive organ of frog ?

(1) Ureter
(2) Ovaries
(3) Urinary bladder
(4) Cloaca

83) Match the following column 'A' and 'B'

	(A)		**(B)**
i	Testes	a	Urinogenital duct
ii	Ureters	b	Ovary
iii	Mesovarium	c	Yellowish ovoid
iv	Cloaca	d	Small median chamber

Choose correct option for reproductive system of frog.
(1) i - c, ii - a, iii - b, iv - d
(2) i - a, ii - c, iii - b, iv - d
(3) i - b, ii - d, iii - a, iv - c
(4) i - b, ii - a, iii - c, iv - d

84) Match the following column with respect to frog :

	Column-I		**Column-II**
A	Urinary bladder	i	cerebellum and medulla oblongata
B	Bidder's canal	ii	kidney of frog
C	Mid brain	iii	bilobed
D	Hind brain	iv	optic lobes

Codes :

	A	**B**	**C**	**D**
1	i	ii	iii	iv
2	iv	iii	ii	i
3	iii	ii	iv	i
4	iii	iv	ii	i

(1) 1
(2) 2
(3) 3
(4) 4

85) **Assertion :** Frog is cold blooded or poikilothermal.
Reason : Frog do not have constant body temperature i.e. their body temperature varies with the temperature of the environment.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

86) Choose incorrect statement ?

(1) Frogs have three chambered heart and are neurogenic in nature.
(2) Main excretory product of frog is urea.
(3) The system for control and coordination is highly evolved and includes both neural system and endocrine glands.
(4) RBC of frog are nucleated and contain red coloured pigments called haemoglobin.

87) The site for final digestion in frog is :

(1) Stomach
(2) Rectum
(3) Oesophagus
(4) Intestine

88) Ureter act as urinogenital duct in :

(1) Male frog
(2) Female frog
(3) In both male and female frog
(4) Cockroach

89) Select the correct route for the passage of sperms in male frog :-

(1) Testes → Vasa efferentia → Kidney → Seminal vesicle → Urinogenital duct → Cloaca
(2) Testes → Vasa efferentia → Ureter → Bidder's canal → Cloaca
(3) Testes → Vasa efferentia → Bidder's canal → Urinogenital duct → Cloaca
(4) Testes → Bidder's canal → Vasa efferentia → Urinogenital duct → Cloaca

90) Which of the following statements is/are correct in reference with the frog ?
A. In water skin act as respiratory organ.
B. Urinary bladder is single lobed in frog.
C. Bidder's canal is present in testes of frog
D. The alimentary canal of frog is short because frogs are omnivores.

(1) Only A
(2) A and C
(3) A, B and C

(4) B and D

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	4	3	2	4	2	1	4	3	1	4	2	1	4	3	4	1	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	3	3	2	3	1	2	3	4	2	1	2	2	4	1	1	2	2	3
Q.	41	42	43	44	45															
A.	4	2	3	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	4	1	2	2	1	1	1	4	3	3	2	3	4	1	3	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	3	2	4	1	1	3	3	1	4	1	3	4	1	1	3	1	4	2	3
Q.	86	87	88	89	90															
A.	1	4	1	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	2	2	3	3	1	1	2	4	4	1	3	2	2	1	4	1	3	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	4	3	4	2	3	3	4	1	1	2	4	4	3	1	3	3	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	2	3	2	4	4	1	3	2	2	1	3	2	4	1	3	2	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	2	3	1	1	3	4	3	4	3	2	2	3	1	2	3	3	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	1	3	1	1	4	1	3	1										

SOLUTIONS

PHYSICS

1) Range $R = \frac{u^2 \sin 2\theta}{g}$ and same for q and 90 – θ
So same for 42° and 48°
Maximum height $H = \frac{u^2 \sin^2 \theta}{2g}$
H is high for higher θ
So H for 48° is higher than H for 42°
Option (2)

2) For speed up angle between $\vec{v}$ and $\vec{a}$ must be acute.

3) $v_y \frac{d}{dt}(y) = \frac{d}{dt}(10t) - \frac{d}{dt}(t^2) = 10 - 2t$
At maximum height, $v_y = 0$
$10 - 2t = 0$
$2t = 10$
$t = 5$ sec.
$y = (10 \times 5 - 5 \times 5)_m = 25$ m

4)

Range are same
$T_1 = \frac{2u \sin \theta}{g}$
$T_2 = \frac{2u \sin(90 - \theta)}{g} = \frac{2u \cos \theta}{g}$
$T_1 T_2 = \frac{4u^2}{g^2} \sin \theta \cos \theta$
$\boxed{T_1 T_2 = \frac{2R}{g}}$

5) Relative displacement between them will be zero at start and end of race.
$S_{rel} = U_{rel.} t + \frac{1}{2} a_{rel.} t^2$
$0 = 4t - \frac{1}{2} \times 1 \times t^2$
on solving
$t = 8$
$xy = 16 \times 8 + \frac{1}{2} \times 1 \times 8^2$
$= 128 + 32 = 160$ m

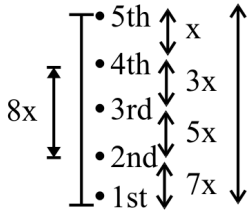
6) $S = 6 + 12t - 2t^2$
$V = 12 - 4t = 0$
$t = 3$
$S_{t=0} = 6$
$S_{t=3} = 6 + 36 - 18 = 24$
$S_{t=5} = 6 + 60 - 50 = 16$
distance = 26m

7) $T = (2u \sin\theta) / g$
$= (2 \times 15 \times 3/5) / 10$
$= 18 / 10$
$= 1.8$ s
Since the man's height is negligible, he must reach the landing point at time T.
Minimum running speed:
v_{min} = distance / time
$= 9 / 1.8$
$= 5$ m/s
Final Answer: Option (2)

8) $x = 1 + 2t^3 + 3t^4$
$v = 6t^2 + 12t^3$
v at $t_2 = 6 \times 4 + 12 \times 8$
$= 24 + 96 = 120$ m/s
v at $t_1 = 6 \times 1 + 12 \times 1$
$= 18$ m/s
$\Delta v = v_f - v_i = 120 - 18 = 102$ m/s

9) $\sum u = 0$
$y = 3x + 5x + 7x$
$\frac{y}{x} = \frac{15}{1}$

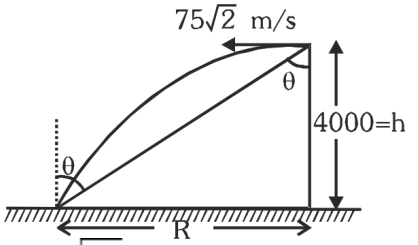
10)



$16x = 48$
$x = \frac{48}{16} = 3$
$8x = 8 \times 3$
$= 24$ m

11)

Let t be the time taken by bomb to hit the target.



$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 4000}{10}} = 20\sqrt{2} \text{ sec}$$

$R = ut = 75\sqrt{2} \times (20\sqrt{2}) = 3000m$

$$\therefore \tan\theta = \frac{R}{h} = \frac{3000}{4000} = 37^\circ$$

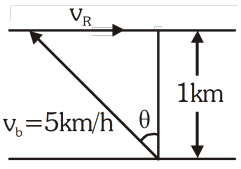
12) $V_A = \tan 53^\circ = \frac{4}{3}$ m/s and

$V_B = \tan 37^\circ = \frac{3}{4}$ m/s

$$3 + \frac{3}{4}t_0 = \frac{4}{3}t_0$$

$$3 = \left(\frac{4}{3} - \frac{3}{4}\right) t_0$$

$$3 = \frac{7}{12}t_0 \Rightarrow t_0 = \frac{36}{7} \text{ sec}$$

13) 

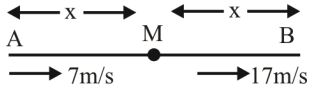
t = 15 min $= \frac{15}{60} = \frac{1}{4}$ hr

$$t = \frac{1}{v_b \cos\theta} = \frac{1}{5\cos\theta} = \frac{1}{4}$$

$$\cos\theta = \frac{4}{5}$$

$$\sin\theta = \frac{3}{5}$$

$$\frac{3}{5} = \frac{v_R}{5} \Rightarrow v_R = 3 kmh^{-1}$$

14) 

$v_m^2 = (7)^2 + 2 \times a \times x$

$(17)^2 = (7)^2 + 2 \times a \times 2x$; $v_m = 13$ m/s

$$\frac{13-7}{a} = t_1 \; ; \; t_2 = \frac{17-13}{a} \; ; \; \frac{t_1}{t_2} = \frac{6}{4} = \frac{3}{2}$$

15)

(a) Scalar can have negative values.

(b) Scalar does not depend on orientation of axes
(c) A vector quantity must follow vector laws.
(d) If the resultant vector is zero then y component of resultant must be zero.

16) $\tan\theta = 2t$

$\sec^2\theta\frac{d\theta}{dt} = 2;$ at $t = \frac{1}{2}$ $\sec \Rightarrow \theta = 45°$

$\frac{d\theta}{dt} = \frac{2}{\sec^2 45^\circ} = 1$

17) Area = $2\int_0^1 x^2 dx = 2\left[\frac{x^3}{3}\right]_0^1 = \frac{2}{3}$

18)

1 m'/s' = 1 × 10 m/20s
= 0.5 m/s
∴ 1 m/s = 2 m'/s'
1 watt' = 1 kg' m'^2/s'^3

$= 1 \times 5\,kg \times \frac{10^2 m^2}{20^3 s^3}$

$= \frac{1}{16} kg\, m^2/s^3$

$\therefore 16\,watt' = 1\,watt$

a & b are correct

19) $m = (v)^a(d)^b(g)^k$
$M = (LT^{-1})^a(ML^{-3})^b(LT^{-2})^k$
a – 3b + k = 0
– a – 2k = 0, b = 1
k = –3
|2k| = 6

20)

$v = \lambda^a g^b \rho^c$
using dimension formula
$\Rightarrow [M^0L^1T^{-1}] = [L^1]^a[L^1T^{-2}]^b[M^1L^{-3}]^c$
$\Rightarrow [M^0L^1T^{-1}] = [M^c L^{a+b-3c} T^{-2b}]$

$\Rightarrow c = 0,\ a + b - 3c = 1,\ -2b = -1 \Rightarrow b = \frac{1}{2}$

Now a + b – 3c = 1

$\Rightarrow a + \frac{1}{2} - 0 = 1$

$\Rightarrow a = \frac{1}{2}$

$\therefore a = \frac{1}{2}, b = \frac{1}{2}, c = 0$

21) Terminal velocity of a spherical body in liquid

$\Rightarrow V_t \propto r^2$

$\Rightarrow \frac{\Delta V_t}{V_t} = 2 \cdot \frac{\Delta r}{r}$

$\Rightarrow \frac{\Delta V_t}{V_t} \times 100\ \% = 2\frac{(0.1)}{5} \times 100 = 4\ \%$

Also $V_t \propto r^2$

Reason R is false

Final Answer: A is true but R is false.

22) Least count (LC) = $\left(\frac{b-a}{b}\right)$ MSD ;

here a = 9,

b = 10 ; 1 MSD = 1 mm

LC = $\left(\frac{10-9}{10}\right)$ 1 mm = 0.1 mm

There is zero error in measurement.

As the zero of the vernier lies to the left of the main scale, it has negative zero error. For negative zero error, the coinciding division is to be read from right and it has 5th division coinciding from right.

Thus zero error = ZE = –5 × 0.1 mm = –0.5 mm = –0.05 cm

Reading at measurement : MSR = 8.5 cm

VSR = 8th VSD × 0.1 mm = 0.08 cm

Corrected reading = MSR+VSR × LC – zero error

= 8.5 cm + 0.08 cm – (–0.5mm) = 8.63 cm

23)

(A) $E = \frac{1}{2}LI^2 \Rightarrow [L] = \frac{[E]}{[I^2]}$

$\Rightarrow [L] = \frac{\left[M^1L^2T^{-2}\right]}{A^2} = \left[M^1L^2T^{-2}A^{-2}\right]$ (A → R)

(B) $E = \frac{1}{2}KT$

$\Rightarrow [K] = \frac{[E]}{[T]} = \frac{\left[M^1L^2T^{-2}\right]}{[K]} = \left[M^1L^2T^{-2}K^{-1}\right]$ (B → D)

(C) $\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow [\mu_0] = \frac{[F]}{[I^2]} = \frac{\left[M^1L^1T^{-2}\right]}{A^2}$

$[\mu_0] = [M^1L^1T^{-2}A^{-2}]$ (C → S)

(D) $R = \rho\frac{\ell}{A} = [\rho] = [RL] = \left[\frac{\rho}{l^2}.L\right]$

$\Rightarrow [\rho] = \frac{\left[M^1L^2T^{-3}L\right]}{A^2} = \left[M^1L^3T^{-3}A^{-2}\right] (D \rightarrow Q)$

24)

For a particular vernier calliper, ZE cancels out when subtracting two measurements from same vernier calliper.

25)

$a = \frac{40}{8} = 5\ m/s^2$

$s = \frac{1}{2}at^2 = 10m$

$\therefore W_g = 40 \times 10 = 400\ J$

26)

Mass = m

Initial linear momentum $\Rightarrow p_i = p$

Final linear momentum $\Rightarrow p_f = p_i + 200\%$ of p_i

$= p + \frac{200}{100} \times p$

$= p + 2p = 3p$

Initial kinetic energy

$\Rightarrow E_{ki} = \frac{p_i^2}{2m} = \frac{p^2}{2m} = E_K(say)$

Final kinetic energy

$\Rightarrow E_{ki} = \frac{p_i^2}{2m} = \frac{(3p)^2}{2m} = 9\left(\frac{p^2}{2m}\right) = 9E_K$

Absolute change in kinetic energy

$\Rightarrow \Delta E_K = (E_{Kf} - E_{Kf}) = (9E_K - E_K) = 8\ E_K$

Percentage change in kinetic energy

$\Rightarrow \frac{\Delta E_K}{E_{ki}} \times 100\% = \frac{8E_k}{E_k} \times 100\% = +800\%$

27) Equilibrium is where the force F is zero.

Stability is determined by the sign of $\frac{dF}{dx}$: Stable if $\frac{dF}{dx} < 0$.

$F = 2x^2 - 3x - 2$

For equilibrium, $F = 0$: $2x^2 - 3x - 2 = 0 \Rightarrow (2x + 1)(x - 2) = 0$.

$x = -\frac{1}{2}$ or $x = 2$.

$\frac{dF}{dx} = 4x - 3$

At x = 2: $\frac{dF}{dx} = 4(2) - 3 = 5 > 0$ (Unstable).

At $x = -\frac{1}{2} : \frac{dF}{dx} = 4\left(-\frac{1}{2}\right) - 3$ $= -2 - 3 = -5 < 0$ (Stable).

28) $H = \frac{v^2}{2g}$

$W_g = -mgH = -mg\left(\frac{v^2}{2g}\right) = \frac{-1}{2}mv^2$

$= \frac{-1}{2}\left(\frac{1}{10}\right)(5)^2$ = –1.25 J

29)

Potential energy of spring $U = \frac{F^2}{2k}$

$\Rightarrow \frac{U_1}{U_2} = \frac{k_2}{k_1} = \frac{3000}{1500} = 2 : 1$

[If F = constant]

30) **Explain the question :** In this question, we need to find how much mass of water (M) a 2 Kw pump can lift to a height of 10 m in 1 minute.

Concept : This question is based on power.

Solution : Power, $P = \frac{\text{work done}}{\text{time}} = \frac{MgH}{t}$

$\Rightarrow M = \frac{P \times t}{gH}$

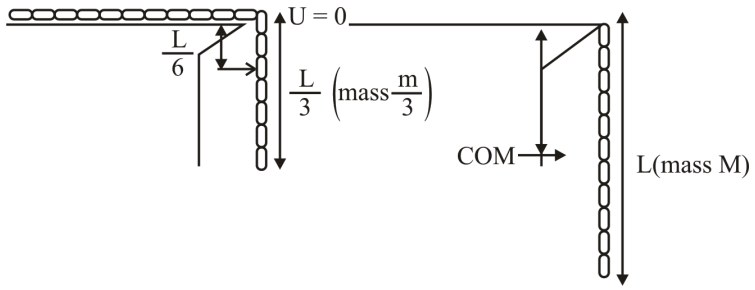
Given here

P = 2kW = 2000 watt, t = 1 min = 60 sec

g = 10 m/s^2, H = 10 M

$\Rightarrow M = \frac{2000 \times 60}{10 \times 10}$ = 1200 kg ≡ 1200 litre (1 litre ≡ 1 kg)

Final answer : Option (4)



31)

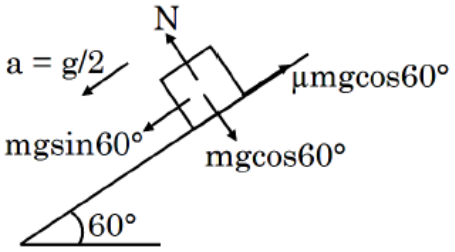
$\boxed{pE_i = -\frac{mg}{3}\frac{L}{6} = -\frac{mgL}{18}}$

$\boxed{pE_f = -mg\frac{L}{2}}$

Now COME law

Gain in KE = Loss in PE

$$= \left(-\frac{mg\ell}{18}\right) - \left(\frac{-mg\ell}{2}\right) = \frac{8mg\ell}{18} = \frac{4mg\ell}{9}$$



32)

$mg \sin 60^\circ - \mu mg \cos 60^\circ = ma$

$g\sin 60 - \mu g \cos 60 = \frac{g}{2}$

$\frac{\sqrt{3}}{2} - \frac{\mu}{2} = \frac{1}{2}$

$\mu = \sqrt{3} - 1$

33) Since, force needed to overcome frictional force.

34) $F = \frac{W}{\mu} = \frac{1 \times 9.8}{0.2} = 49N$

35) The sand bag is placed on a trolley that is moving with a uniform speed of 27 km/h. The external forces acting on the system of the sandbag and the trolley is zero. When the sand starts leaking from the bag, there will be no change in the velocity of the trolley. This is because the leaking action does not produce any external force on the system. This is in accordance with Newton's first law of motion. Hence, the speed of the trolley will remain 27 km/h.

36) Weight of the disc will be balanced by the force applied by the bullet on the disc in vertically upward direction.

F = 2nmv = Mg

$= 2 \times 20 \times 0.05 \times 6 = M(10)$

$M = \frac{2 \times 20 \times 0.05 \times 6}{10} = 1.2$ kg

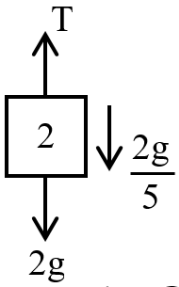
37) Due to friction (a = μg) velocity of block will become equal to velocity of belt. Relative motion between two will stop.

v = at = μgt

= 0.2 × 10 × 4 = 8 m/sec

38)

$a = \frac{2g}{2+3} = \frac{2g}{5}\ m/s^2$



$T = M\left(g - \frac{2g}{5}\right) = \frac{6g}{5}N$

39)

$F_s = (2.5g \cos 37°) \sin 37°$

$= 25 \times \frac{3}{5} \times \frac{4}{5} = 12N$

40) Force produced by the engine $F = \frac{P}{v} = \frac{30 \times 10^3}{30} = 10^3 N$

$$\text{Acceleration} = \frac{\text{Forward force by engine-resistive force}}{\text{mass of car}}$$

$$= \frac{1000 - 750}{1250} = \frac{250}{1250} = \frac{1}{5} m/s^2$$

41) $W = -\int_{20}^{30} F dx$

$W = -\int_{20}^{} 0.1x\, dx$

$W = -0.1\left[\frac{x^2}{2}\right]_{20}^{30}$

$W = -0.1\left[\frac{900 - 400}{2}\right] = -25$ J

From work energy theorem $W = K_f - K_i$

$\Rightarrow \quad -25 = K_f - \frac{1}{2} 10(10)^2$

$\Rightarrow \quad K_f = 475$ J

42) $\because U = \sqrt{2gH} = \sqrt{32 \times 2g} = \sqrt{64g}$

$\because t = \frac{U}{g} = \frac{\sqrt{64g}}{g}$

$\therefore 4 = \sqrt{\frac{64}{g}} \quad \therefore g = 4 m/s^2$

$\because S_{n^{th}} = U + \frac{a}{2}(2n - 1)$

$\therefore S_{4^{th}} = \sqrt{64 \times 4} - \frac{4}{2}(2 \times 4 - 1)$

= 16 – 14 = 2 m

43) Under area of F – t = ΔP

so, P will be maximum at t = 15 sec.

44) $\frac{dw}{dt} = P = 2x$

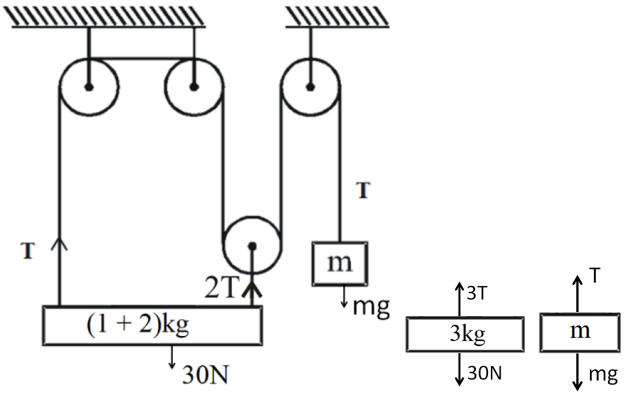
FV = 2x

$\left(mv\frac{dv}{dx}\right)(v) = 2x$

$\int_0^v v^2\,dv = \left(\frac{2}{m}\right)\int_0^x x\,dx$

$\frac{v^3}{3} = \frac{2}{m}\frac{x^2}{2}$

$v = \left(\frac{3}{m}x^2\right)^{1/3}$



45)

3T = 30 .... (1)

T = mg .... (2)

Put T in equation ---------- (1)

3mg = 30

m = 1kg

CHEMISTRY

46)

(L.R.)

$3A + 2B \rightarrow C + \frac{1}{2}D$

12 mol 12 mol

Moles of D = $\frac{1}{6} \times 12 = 2$

$\frac{60}{M_w} = 2 \Rightarrow M_w = 30$

47) NCERT Pg # 25

$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O(g)$

moles of $C_2H_6 = \frac{3}{30} = 0.1$

moles of $H_2O(g) = 0.1 \times 3 = 0.3$

vol of $H_2O(g)$ at STP = $0.3 \times 22.4 = 6.72$ L

48)

Let the mass of pure $CaCO_3$ in the sample be x g

According to the reaction

100 g $CaCO_3 \rightarrow$ 56 g CaO

So,

$(56/100) \times x = 56$

$\Rightarrow x = 100$g

% purity = $(100/120) \times 100 = 83.3\%$

Final Answer:

Percentage purity of $CaCO_3$, = 83.3%

Correct Option: (2) 83.3%

49)

Number of molecules = $1.4 \times N_A = 1.4 \times 6.022 \times 10^{23} = 8.43 \times 10^{23}$

Total atoms = $8.43 \times 10^{23} \times 4 = 3.372 \times 10^{24}$

Final Answer :

Option (4)

50) **Concept:** Mole relation: 1 mole of CO_2 has 2 O atoms; 1 mole of CO has 1 O atom.

Solution: Molar mass of $CO_2 = 12 + (2 \times 16) = 44$ g/mol

Moles of CO_2 in 22g:

$n(CO_2) = 22/44 = 0.5$ mol

Moles of O atoms in $CO_2 = 0.5 \times 2 = 1$ mol O atoms

Each CO molecule has 1 O atom → need 1 mol CO

Mass of 1 mol CO = 12 + 16 = 28 g

Final answer: option (1)

51) Equivalent mass of metal = $\frac{\text{mass of metal}}{\text{mass of oxygen}} \times 8$

$= \frac{68}{32} \times 8 = 17$

52) $$\% = \frac{\text{Atomic weight} \times (\text{no. of atoms} = 1)}{\text{minimum molecular weight}} \times 100$$

$$0.4 = \frac{39 \times 1}{(M_w)_{min.}} \times 100$$

$$(M_w)_{min} = \frac{39}{0.40} \times 100 = 9750$$

53) $C_{\frac{38.8}{12}}H_{\frac{16}{1}}N_{\frac{45.2}{14}} \Rightarrow C_{3.23}H_{16}N_{3.23}$

$= C_1H_5N$

$= CH_3NH_2$

54)

no. of moles of Co_2 left = $n_2 - n_1$

$$= \frac{200 \times 10^{-3}}{44} - \frac{10^{21}}{6.02 \times 10^{23}}$$

$$= 2.88 \times 10^{-3} \text{moles}$$

Final Answer - option (1)

Level - Medium

55)

$H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$

H_3PO_4 donates **3 H^+ ions**

Valency factor of $H_3PO_4 = 3$

Molecular weight of $H_3PO_4 = (3\times1) + 31 + (4\times16) = 98$

Equivalent weight = $98 \div 3 = 98/3$

FINAL ANSWER: Option (1)

56) $\underset{+6}{\underline{Cr_2O_7^{2-}}} + \underset{+4}{\underline{SO_3^{2-}}} \rightarrow \underset{+3}{\underline{Cr^{3+}}} + \underset{+6}{\underline{SO_4^{2-}}}$

$8H^+ + Cr_2O_7^{2-} + 3SO_3^{2-} \rightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O$

no. of Cr^{3+} : no. of $H^+ = 2 : 8$

$= 1 : 4$

Final answer: Option (4)

57) Oxidation number of S in H_2SO_4 is +6 which is maximum. hence it can not further oxidise and can not behave as reducing agent.

58) Oxidising agent has a tendency to gain electrons

59)

Question Explanation: To Calculate equivalent weight of H_3PO_2 disproportionation reaction.

Concept: Equivalent weight in a redox reaction.

Solution:

$$\overset{+1}{H_3PO_2} \longrightarrow \overset{-3}{PH_3} + \overset{+3}{H_3PO_3}$$

H_3PO_2 disproportionate , So v.f while getting oxidised = 3 – 1 = 2

v.f while getting reduced = 1 – (–3) = 4

Overall v.f (n)

$$\frac{1}{n} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$n = \frac{4}{3}$$

$$\text{Wet} = \frac{M \times 3}{4} = \frac{3M}{4}$$

Final Answer: Option (2)

60) **Question Explanation:**

To calculate mass of $K_2Cr_2O_7$, required to oxidise Mohr's salt.

Concept:

Low of chemical equivalence.

Solution:

$$\overset{+2}{F}eSO_4(NH_4)SO_4.6H_2O + K_2\overset{+6}{K_2}Cr_2O_7 \rightarrow F_e^{3+} + Cr^{3+}$$

v.f of Mohr's salt = 1 (iron changing from + 2 to + 3)

v.f of $K_2Cr_2O_7 = 3 \times 2 = 6$

eq. of Mohr's salt = Gm eq. of $K_2Cr_2O_7$

$$\frac{750}{1000}L \times 0.6 \times 1 = \text{moles} \times 6$$

$$\Rightarrow \text{moles of } K_2Cr_2O_7 = \frac{750 \times 0.6}{1000 \times 6}$$

$$\text{wt of } K_2Cr_2O_7 = \frac{750 \times 0.6}{1000 \times 6} \times [39 \times 2 + 52 \times 2 + 16 \times 7]$$

$$= \frac{750 \times 0.6}{1000 \times 6} \times 294 = 22.05g$$

Final Answer:

Option (3)

61)

$$N_2H_4 \rightarrow 2NO_3^-$$

$$6H_2O + N_2H_4 \rightarrow 2NO_3^- + 16H^+ + 14e^-$$

62) $K = \frac{(0.4)^2}{(0.1)} \Rightarrow 1.6$ M

63)

$$K_C = \frac{K_f}{K_b} = \frac{K_f}{2500\,K_f} = \frac{1}{2500}$$

$$K_p = K_C(RT)^{\Delta n_g} = \frac{1}{2500}(0.0831 \times 1000)^1$$

$K_p = 0.033$

64)

	PCl_5 (g)	⇌	PCL_3 (g)	+	Cl_2 (g)
t = 0	2		–		–
t_{eq}	2 – x		x		x

%a = 35% ; $\frac{35}{100} = \frac{x}{2}$ ⇒ x = 0.7

Total number of moles at equilibrium = 2 – x + x + x

= 2 + x

= 2 + 0.7 = 2.7

Final answer: 4

65)

Explain the question

Analyze ice-water equilibrium shifts.

Concept

Le Chatelier's Principle.

Solution

For the equilibrium $H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$, the forward process (melting) is endothermic ($\Delta H>0$); thus, increasing temperature shifts the equilibrium forward. Regarding pressure, water exhibits anomalous behavior where the density of ice is less than liquid water, meaning the volume of ice is greater than the volume of water ($V_{ice} > V_{water}$). According to Le Chatelier's Principle, increasing pressure shifts the equilibrium toward the side with lower volume (the liquid phase). While the Assertion is true because both factors favor water formation, the Reason is false because it incorrectly claims the volume of water is greater than the volume of ice.

Final answer : Option 2

66)

Explain the Question : We need to find relation between two equilibrium constants for given two reactions.

Concept : This question is based on Factors affecting K_{eq}.

Solution :

$2A + B \rightleftharpoons 2C$; $K_{eq} = 40$

↓ Reaction is reversed

$2C \rightleftharpoons 2A + B$; $K_{eq} = \frac{1}{40}$

↓ Multiplied by $\frac{1}{2}$

$C \rightleftharpoons A + \frac{1}{2}B$; $K_{eq} = \left(\frac{1}{40}\right)^{\frac{1}{2}}$

Keq = $\sqrt{\frac{1}{40}}$

Final answer : 3

67) **Explain the question :** We need to find direction of reaction if value of K_C and concentration of reactants and product are given.

Concept : This question is based on reaction quotient.

Solution :

$2A \rightleftharpoons B + C$; $K_c = 4 \times 10^{-3}$

At a given time ⇒ [A] = [B] = [C] = 2×10^{-3} M

$Q_c = \frac{[B][C]}{[A]^2}$

$Q_c = 1$

as $Q_c > K_c$

Reaction has a tendency to go in backward direction.

Final answer : 3

68)

Explain the Question : We need to find relation between K_P and total pressure (P) for reaction if % α is given.

Concept : This question is based on K_P involving degree of dissociation.

Solution : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

t = 0	1	–	–
t_{eq}	1– 0.5	0.5	0.5

= 0.5

$(P)_{PCl_5} = X_{PCl_5} \cdot P_T = \frac{0.5P}{1.5} = \frac{P}{3}$

$(P)_{PCl_3} = X_{PCl_3} \cdot P_T = \frac{0.5P}{1.5} = \frac{P}{3}$

$(P)_{Cl_2} = X_{Cl_2} \cdot P_T = \frac{0.5P}{1.5} = \frac{P}{3}$

$K_P = \frac{\left(\frac{P}{3}\right)^2}{\left(\frac{P}{3}\right)} \Rightarrow K_P = \frac{P}{3} \Rightarrow P = 3K_P.$

Final answer : 2

69)

All the measurable properties of the system are variable at equilibrium state:
Incorrect: At equilibrium, all measurable properties are constant. They do not change with time.
Conclusion:
The statement that is not true is "All the measurable properties of the system are variable at equilibrium state."
Hence,
Option (4) is correct

70) • The reaction is exothermic because heat (242 kcal) is a product. To favour the forward reaction, the temperature must be lowered.
• **Pressure**: Increasing pressure favours the side with fewer gas moles.
• Since the product side has fewer moles of gas, high pressure will favour product formation.
The correct conditions are low temperature and high pressure

Final Answer: Option 1.

71) The formula for the degree of hydrolysis (h) for a WAWB salt is:

$$h = \sqrt{\frac{K_\omega}{K_a.K_b}}$$

• In this formula, three is no term for concentration (C).
• Therefore, whether the concentration is 0.001 M or 0.002 M, the value of h (represented here as x) remains the same.
• Result: $x_1 = x_2$.

Final Answer: Option (1)

72) $N_1V_1 = N_2V_2$
$0.2 \times 2 \times V = 0.1 \times 2 \times 5$
$V = \frac{5}{2} = 2.5$ L

73) **Q. Explanation :** We need to find highest degree of hydrolysis(h).
Concept : Degree of hydrolysis.
Explanation :
$h = \sqrt{\frac{Kw}{C \times Kb}}$
As concentration decreases, the degree of hydrolysis increases.
0.001 M NH_4Cl → Lowest concentration = Highest hydrolysis
Answer: option 3, 0.001 M NH_4Cl

74)

Explain the Question : We are asked why an aqueous solution of sodium carbonate (Na_2CO_3) is alkaline.

Concept : This question is based on hydrolysis of WASB
Solution :
Na_2CO_3 is a salt of a strong base (NaOH) and a weak acid (H_2CO_3).
In water, Na^+ ions do not hydrolyze (they come from a strong base).
CO_3^{2-} ions undergo hydrolysis :
$CO_3^{2-} + H_2O \rightleftharpoons 2OH^- + H_2CO_3$
This reaction produces OH^- ions, making the solution alkaline.
Final answer : 1

75)

Explain the Question : Need to find pH of the solution buffer solution .
Concept : This question is based on buffer solution.
Solution : $\left.\begin{matrix} NH_4Cl & + & NH_4OH \\ \text{salt} & & \text{weak base} \end{matrix}\right\}$ make basic buffer .

$$pOH = pKb + \log \frac{[salt]}{[base]}$$

$$pOH = 5 + \log \left(\frac{10^{-1}}{10^{-2}}\right)$$

$pOH = 5 + \log 10$
$pOH = 5 + 1 = 6$
$pH = 14 - 6 = 8$
Final answer : 4

76) **Case-I**

$NaOH$	+	NH_4Cl	→	$NaCl$	+	NH_4OH
1 mol		1 mol		0		0
0		0		1 mol		1 mol
				(Neutral)		

$$[OH^-]_{NH_4OH} = \sqrt{K_b \times C}$$

$$(pOH)_1 = \frac{1}{2} pK_b \quad \left(C = \frac{1 \text{ mol}}{1L} = 1\right)$$

Case II
$NH_4OH + NH_4Cl \Rightarrow$ Basic Buffer
1 mol 1 mol

$$(pOH)_{II} = pK_b + \log \frac{[NH_4Cl] = 1}{[NH_4OH] = 1} = pK_b$$

$$\frac{(pOH)_I}{(pOH)_{II}} = \frac{1/2\ pK_b}{pK_b} = \frac{1}{2}$$

77)

Concept : This question is based on
Precipitation occurs when : $K_{sp} = [Pb^{2+}]\ [OH^-]^2$
Relation between pH and $[OH^-]$: $pH + pOH = 14$, $[OH^-] = 10^{-pOH}$
Solution :

$K_{sp} = [Pb^{2+}][OH^-]^2$
$1.2 \times 10^{-5} = 0.12 \times [OH^-]^2$
$[OH^-]^2 = \frac{1.2 \times 10^{-5}}{0.12} = 1.0 \times 10^{-4}$
$[OH^-] = 10^{-2}$ M
pOH = 2 ⇒ pH = 14 – 2 = 12
Final answer : 3

78)

Concept : This question is based on $\Delta S = \sum S_{products} - \sum S_{reactants}$
Solution :
Entropy of products = 2 × S(HCl) = 2 × 186.7 = 373.4 J K^{-1} mol^{-1}
Entropy of reactions = $S(H_2) + S(Cl_2)$ = 130.6 + 223 = 353.6 J K^{-1} mol^{-1}
ΔS = 373.4 – 353.6 = + 19.8 J K^{-1} mol^{-1}
Final answer : 4

79) **Question Explanation :**
Question is asking about Match the sign of ΔH and ΔS to the nature of the reaction.

Concept :
This question is based on spontaneity of reaction.
Solution :
(P) Spontaneous only at low temperature :-
ΔH = –ve, ΔS = –ve, ΔG = –ve (A–P)
(Q) Spontaneous only at high temperature :-
ΔH = +ve, ΔS = +ve, ΔG = –ve (C–Q)
(R) Spontaneous only at all temperature :-
ΔH = –ve, ΔS = +ve, ΔG = –ve (B–R)
(S) Spontaneous only at all temperature :-
ΔH = +ve, ΔS = –ve, ΔG = +ve (D–S)
Final Answer :
Option (1)

80)

Concept:

A. **Adiabatic Process:** An adiabatic process is one in which no heat is exchanged between the system and the surroundings (q = 0).
B. **Reversible Process:** A reversible process is one that can be reversed by an infinitesimal change in a property of the system.

Mathematical Calculation:

For a reversible adiabatic process, the change in entropy (ΔS) is always zero. This is because there is no heat transfer (q = 0) and the process is reversible.

ΔS = q(rev)/T. Since q = 0, then ΔS = 0.

Final Answer:

The change in entropy (ΔS) is zero.

Therefore, the correct answer is option 1: Zero.

81)

First Law of Thermodynamics:
$\Delta U = q + w$
$0 = 208\ J + w$
$w = -208\ J$
Final Answer
The values are q = +208 J and w = -208 J.
Therefore, the correct answer is (3) q = +208 J, w = –208 J.

82) Combustion reaction :-
$CH_4(g) + 2CO_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
$\because \Delta ng = 1 - 3 = 2$
$\Delta H = -800 + \left(-2 \times \frac{8.314 \times 298}{1000}\right)$
$\Delta H = -805\ KJ$
Final Answer :
Option (1)

83)

$C + O_2 \rightarrow CO_2$; $\Delta_c H = -390\ kJ\ mol^{-1}$
1 mole = 44 g → 390 kJ released
For 35 g → $\frac{390}{44} \times 35 = 310\ kJ$

84) For 1 mole HCl and NaOH; Heat = –57.3 kJ
For 0.25 mole; Heat = –57.3 × 0.25
= –14.32 kJ
(Unit should not be kJ mol^{-1})

85) $\Delta H^o_{comb}(H_2) = \Delta H^o_f(H_2O)$
Calorific value = $\frac{\Delta H^o_{comb}(H_2)}{Mw_{H_2}}$
$-143 = \frac{\Delta H^o_f(H_2O)}{2}$
$\Delta H^o_f(H_2O) = -286\ kJ\ mol^{-1}$

86)

Target equation = (eq. (ii) × 2) + eq. (iii) – eq.(i)
ΔH = (–393.5 × 2) + (–285.8) – (–1299.5)
ΔH = + 226.7 kJ

87) $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightarrow NO(g)$
Desired equation $= \frac{1}{2}x(1) - \frac{1}{2}x(2)$
$\Delta H = \frac{1}{2}(x - y)$

88) NCERT-XI, Pg. # 172
$\frac{1}{2}H_{2(g)} + \frac{1}{2}X_{2(g)} \rightarrow HX_{(g)}\ \Delta H = -50kJ$
$\Delta H = \Sigma(B.E.)_{Reactants} - \Sigma(B.E.)_{Products}$

89) $C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$
ΔH^0 = 7(–394) + 8(–286) – (–200)
ΔH^0 = –2758 – 2288 + 200 = –4846

90) **Question Explanation**
To compare Q_1 and Q_2
Concept:
Laws of thermochemistry
Solution:
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O + Q_1$
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell) + Q_2$
liquidis more stable and have more force of attraction so in case of H_2O (□) formation more heat is released .
So, $Q_2 > Q_1$

Final answer: Option (3)

BIOLOGY

91) NCERT XI Page No. # 1.31/9

92)

Module
The correct answer is C & D

93)

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94)

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95)

Module

96)

NCERT-XI, Pg. # 11

97)

NCERT-XI, Pg. # 10

98)

NCERT-XI, Pg. # 90, 91

99)

NCERT-XI, Pg. # 14

100) NCERT XI Pg. # 13

101) NCERT XI Pg. # 19,20

102) NCERT Pg. No. 20, 21, 22, 2.2 - 2.2.1, 2.2.2, 2.2.3, 2.2.4, 2.2.5

103)

NCERT-XI, Pg. # 14, 15

104)

NCERT -XI, Pg. # 14, 15
Statements A, B, D and E are correct.

105)

NCERT-XI, Pg. # 14, 15

106)

NCERT XI, Pg.#23, 24

107) NCERT Pg. # 18

108)

NCERT-XI, Pg. # 17-18

109)

NCERT-XI, Pg. # 16, 17

110)

NCERT-XI, Pg. # 16

111)

XI NCERT Pg. No. # 24

112)

NCERT-XI, Pg. # 19, 20, 21

113)

NCERT-XI, Pg. # 19, 20

114)

NCERT-XI, Pg. # 20

115)

NCERT-XI, Pg. # 21

116)

NCERT-XI, Pg. # 12 to 18

117) NCERT-XI, Pg. # 20-23, 27

118)

NCERT-XI, Pg. # 19, 20, 21

119)

NCERT-XI, Pg. # 26, 27, 28

120) NCERT-XI, Pg. # 27, 28
Pheophyceae (Brown algae) :-
Fucus, Dictyota, Sargassum
Rhodophyceae :-
Gelidium, Porphyra, Polysiphonia, Chondrus

121) NCERT-XI, Pg. # 30

122)

NCERT Pg. No. 42

123) NCERT Pg # 34

124) NCERT-XI, Pg. # 29

125) NCERT No. Pg.#29

126)

NCERT-XI, Pg. # 32

127)

NCERT-XI, Pg. # 29, 31, 33, 34

128)

NCERT-XI, Pg. # 32

129)

NCERT-XI, Pg. # 33

130) NCERT-XI, Pg. # 31

131) NCERT XI Pg.#33,34

132)

NCERT-XI, Pg. # 32, 33, 34
Selaginella, Cycas, Pinus, Eucalyptus, Salvinia

133)

NCERT-XI, Pg. # 26

134) NCERT-XI, Pg. # 32

135)

NCERT-XI, Pg. # 26 ; Table 3.1

136) NCERT Pg. # 43

137) NCERT Pg. # 45

138) NCERT Pg. # 45

139)

They are covered by ctenoid/cycloid scales.

140)

Pg. No. # 45

141)

Bath sponge is common name of Euspongia.

142)

NCERT, Pg. # 45-47

143) NCERT–XI, Pg. # 56 (Fig. 4.18)

144) NCERT-XI, Pg. # 47-48

145) NCERT Pg.#42-45.

146) NCERT-XII, Pg. # 104

147) NCERT-XI, Pg. # 45, Para-4.2.9

148)

NCERT, Pg. # 47-48

149) NCERT (XIth) Pg. # 53, IInd - Para

150) NCERT XI Pg # 58

151) NCERT XI Pg.#55

152) NCERT, Pg # 47-48

153) NCERT, Pg. # 39

154)

Module

155)

Module

156) **Question Explanation :**
The question is asking to correctly match type of connective tissues in Column I with their respective characteristics or cell types in Column II.
Concept:
This question is based on connective tissue.
Solution:
A – Areolar tissue → Loose connective tissue
B – Adipose tissue → Contains fat cells
C – Ligaments → Mode of dense regular connective tissue
D – Bone → Contains osteocytes.
Final Answer: Option (1)

157) Module

158)

Module

159)

NCERT XI Pg # 102, Para 7.1.1

160) NCERT Page No. 101,102

161) NCERT XI Pg. 111

162)

A-Ocellus, B-Compounds eye, C-Mandible, D-Maxilla, E-Labrum, F-Labium

163) NCERT Page - 111 and 112

164) NCERT Page - 112 and 114

165) NCERT Pg.#112 (E), 113 (H)

166) NCERT Pg.#114 (E), 114 (H)

167)

NCERT 11th Pg. No. 112

168)

Old NCERT, Pg. # 112

169) NCERT Pg. # 114

170)

NCERT, Pg. # 82

171) NCERT Pg. No.116/7.5.1

172) NCERT Pg. No.119/7.5.2

173) NCERT Pg. No.119/7.5.2

174) NCERT Page - 118 and 119

175) NCERT Pg. # 116

176) NCERT-XI, Pg. # 118

177)

NCERT, Pg. # 81-82

178)

In male frogs, the ureters serve a dual function as urinogenital ducts. They carry both urine from the kidneys and sperm from the testes to the cloaca, where it is expelled from the body.ans-1

179)

NCERT, Pg. # 83

180) NCERT Page - 116 to 119